Sri Lanka Institute of Information Technology

Integrated Furniture Shop Management System

Project Report

Information Systems Project 2026

Project ID: ISP_G08

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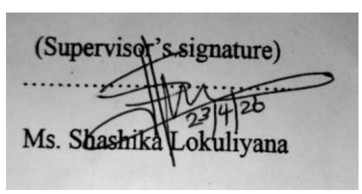

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Abstract

The Integrated Furniture Shop Management System was developed to address operational inefficiencies associated with traditional manual management practices in furniture retail environments, such as inaccurate inventory tracking, delayed order processing, fragmented payment handling, and limited delivery coordination visibility. These challenges often reduce service efficiency and affect customer satisfaction. To overcome these limitations, a centralized web-based management system was designed and implemented using the Spring Boot framework, Java programming language, HTML-based user interfaces, and a MySQL relational database. The system integrates key operational modules including user and account management, product management, shopping cart and order processing, payment and billing, delivery tracking, and customer feedback handling within a unified platform. The implemented solution enables administrators to manage business operations more efficiently while allowing customers to browse products, place orders securely, and monitor delivery progress in real time. System testing confirmed that the application successfully supports core operational workflows with improved accuracy and coordination compared to manual processes. Overall, the developed system enhances business efficiency, reduces processing errors, improves transaction transparency, and provides a scalable digital foundation for future enhancements such as mobile platform integration and advanced business analytics capabilities.

Acknowledgement

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Special thanks go to all team members for their cooperation, dedication, and commitment throughout the development of the Furniture Shop Management System. We would like to thank everyone who supported us directly or indirectly during the completion of this project.

Declaration

We declare that this project report or part of it was not a copy of a document done by any organization, university any other institute or a previous student project group at SLIIT and was not copied from the Internet or other sources.

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Project ID	ISP_G08

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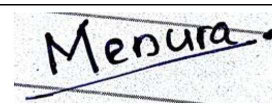
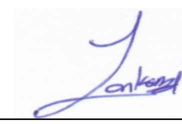
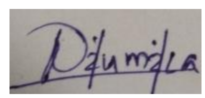
IT24100885	Perera A.J.M..P	
IT24100696	Dissanayake L. N	
IT24101035	Sewwandi R.P.D.D.C	
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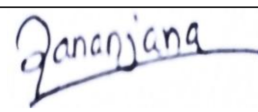
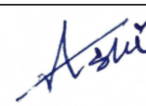
IT24101357	Bandara N.H.M.W.W.D.S	
IT24100863	Ananda U.V.D.M.A	

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List of Acronyms and Abbreviations

- UML - Unified Modeling Language
- HTTPS - Hypertext Transfer Protocol Secure
- SQL – Structured Query Language
- REST - Representational State Transfer
- HTML - Hypertext Markup Language
- EER - Enhanced Entity Relationship
- PDF – Portable Document Format
- DBMS – Database Management System
- JDBC – Java Database Connectivity

1. Introduction

1.1 Problem Statement

The furniture retail business sector often relies on manual methods for managing inventory, customer orders, payments, deliveries, and customer support. In the selected furniture shop, most operational activities are maintained using paper-based records and spreadsheets, resulting in inaccurate inventory tracking, delayed order processing, difficulties in monitoring deliveries, and limited visibility of business transactions. These issues reduce operational efficiency and increase the risk of human error.

To overcome these challenges, a centralized Integrated Furniture Shop Management System is proposed to automate inventory management, order processing, payment handling, delivery tracking, and customer support activities. The system provides real-time access to business information, improves accuracy and coordination across operations, and supports better decision-making while enhancing overall customer service quality.

1.2 Product Scope

1.2.1 Overview of the Proposed System

The Integrated Furniture Shop Management System is a centralized web-based platform designed to automate core operational activities of a furniture retail business environment. The system replaces manual record-keeping processes related to inventory tracking, order processing, payment handling, delivery coordination, and customer support with an integrated digital solution that improves operational efficiency and data accuracy.

Administrators manage product records, monitor orders, coordinate deliveries, and respond to customer requests, while customers can browse products, place orders, make secure payments, track deliveries, and submit feedback through the system.

1.2.2 Purpose of the System

The purpose of the system is to improve workflow automation by providing real-time inventory visibility, structured order processing, secure transaction handling, efficient delivery coordination, and organized customer support management. The system also supports business performance monitoring through digital transaction tracking and reporting facilities.

1.2.3 System Scope Boundaries

The system supports the following operational modules:

1. User Management
2. Product Management
3. Shopping Cart and Order Processing Management
4. Payment and Billing Management
5. Delivery Management
6. Feedback and Support Management

1.2.4 Project Objectives

1.2.4.1 Main Objective

To design, develop, and deploy a centralized Integrated Furniture Shop Management System within a 12-week development period that supports inventory management, product management, customer order processing, secure payment handling, delivery management, and customer support services, while reducing manual operations and improving workflow efficiency.

1.2.4.2 Sub-Objectives

Table 1.1 : Sub Objectives

Module	Sub-objective
User Management	Provide secure registration, authentication, password recovery, and profile management for customers and administrators.
Product Management	Enable administrators to manage product information, categories, pricing details, and inventory records.
Shopping Cart & Order Management	Allow customers to manage shopping carts, place orders, and track order status throughout the order lifecycle.

Payment & Billing Management	Support secure payment processing, invoice generation, and transaction history management.
Delivery Management	Facilitate delivery scheduling, assignment, and real-time delivery status tracking.
Feedback & Support Management	Enable customers to submit reviews and support requests while allowing administrators to manage responses efficiently.

1.2.5 Benefits of the Proposed System

The proposed system improves inventory accuracy, enhances transaction reliability, supports delivery coordination, and strengthens customer communication through a centralized digital platform. It also provides structured business information that supports decision-making and improves overall service quality within the furniture shop environment.

1.2.6 Alignment with Organizational Goals and Business Strategy

The system supports the organization's digital transformation by improving operational efficiency, reducing manual errors, enhancing customer service transparency, and providing accurate business information for decision-making and operational planning.

1.2.7 Summary of Key Findings from Requirements Analysis

Requirement's analysis identified that manual inventory handling, delayed order processing, limited delivery tracking visibility, and lack of centralized transaction monitoring were major operational weaknesses in the existing system environment. These findings guided the identification of six functional modules that form the foundation of the proposed system design.

1.3 Project Report Structure

This project report is organized into four chapters .

Chapter 1 - presents the problem statement, product scope, and objectives of the proposed system.

Chapter 2 - describes the methodology including requirements analysis, system design, implementation approach, and testing activities.

Chapter 3 - evaluates the system results and discusses lessons learned and future improvements.
Chapter 4 - concludes the report by summarizing project outcomes and organizational benefits.
The report also includes references prepared using IEEE format and appendices containing supporting design diagrams, test results, and implementation details.

2. Methodology

2.1 Requirements and Analysis

The requirements and analysis phase focused on identifying business needs, understanding existing operational limitations, and defining functional and non-functional system requirements necessary for developing a centralized digital solution for furniture retail management. Multiple requirements elicitation techniques were used to ensure that system requirements accurately reflected stakeholder expectations and operational workflows.

2.1.1 Requirements Elicitation

Table 2.1 : Requirements Elicitation

Technique	Purpose
Stakeholder Interviews	Identify operational challenges and business requirements.
Observation	Examine existing inventory, order processing, and delivery workflows.
Document Analysis	Review inventory records, order forms, payment receipts, and delivery documents.
Brainstorming	Refine requirements and identify additional system features.

2.1.2 Stakeholder Requirements Analysis

Based on the collected information, key stakeholders of the system were identified as the furniture shop owner, administrators, customers, delivery manager, finance staff, and customer support officers. Each stakeholder group interacts with the system differently and contributes to specific operational workflows such as product management, order processing, transaction monitoring, delivery scheduling, and service feedback handling. Understanding stakeholder roles helped ensure that the proposed system supports both administrative management activities and customer interaction processes within a unified operational platform.

2.1.3 Functional Requirements

Table 2.2 : Functional Requirements

Module	Description
User and Account Management	Supports registration, login, authentication, password recovery, and profile management.
Product Management	Enables product creation, updating, categorization, and inventory maintenance.
Shopping Cart and Order Management	Supports cart management, order placement, order confirmation, and order tracking.
Payment and Billing Management	Supports secure payment processing, invoice generation, and transaction management.
Delivery Management	Supports delivery scheduling, delivery tracking, and status updates.
Feedback and Support Management	Enables customer reviews, support ticket creation, and issue resolution tracking.

2.1.4 Non-Functional Requirements

In addition to functional capabilities, several non-functional requirements were identified to ensure system reliability, usability, security, and performance during operation.

Table 2.3 : Non-Functional Requirements

NFR	Description
Security	Secure authentication and encrypted transaction processing to protect user information.
Performance	Efficient page loading and support for multiple concurrent users.
Reliability	Accurate transaction processing and data storage without information loss.
Usability	User-friendly interfaces for both administrators and customers.
Availability	Continuous system accessibility except during scheduled maintenance periods.
Maintainability	Easy system updates, configuration management, and future enhancements.

2.1.5 System Constraints

Several constraints were identified during the requirements analysis phase that influenced system design and implementation decisions. The system must integrate with a secure third-party payment gateway to support encrypted financial transactions. Development activities were conducted following the Agile Scrum methodology to ensure structured sprint-based implementation. In addition, the system was designed to integrate with inventory management workflows and delivery coordination processes within the furniture shop environment.

2.1.6 Use Case and Activity Diagram Support for Requirements Analysis

To support the analysis of functional requirements, use case specifications and UML use case diagram(figure 2.7) were developed during the requirements engineering phase. These diagrams illustrate the interactions between system actors and the services provided by the Integrated Furniture Shop Management System. Figures 2.1 to 2.6 present detailed use case specifications for the major functional modules identified during requirements elicitation.

E1 – User & Account Management

Use Case	Register Account
Primary Actor	Customer
Preconditions	• Customers have access to the system. • Customer is not already registered.
Main Flow	1. Customer opens the system registration page. 2. Customer enters personal details (name, email, password). 3. System validates the input fields. 4. System creates the customer account. 5. System sends an email verification link.
Postconditions	Customer account is successfully created and stored in the system.

Figure 2.1 : User Management Use Case

E2 – Product Management

Use Case	Add Product
Primary Actor	Admin
Preconditions	• Admin is logged into the system.
Main Flow	1. Admin navigates to the product management section. 2. Admin selects **Add Product**. 3. Admin enters product details (name, category, price, description). 4. Admin uploads product images. 5. System validates the information. 6. System saves the product in the database.
Postconditions	New furniture product becomes available in the system.

Figure 2.2 : Product Management Use Case

E3 – Shopping Cart & Order Management

Use Case	Place Order
Primary Actor	Customer
Preconditions	• Customer is logged in. • Customer has added items to the cart.
Main Flow	1. Customer opens the shopping cart. 2. Customer reviews selected items. 3. Customer confirms the order. 4. System validates product availability. 5. System creates the order record. 6. System displays order confirmation.
Postconditions	Order is successfully recorded in the system.

Figure 2.3 : Shopping Cart and Order Management Use Case

E4 – Payment & Billing

Use Case	Process Payment
Primary Actor	Customer
Preconditions	• Customer has placed an order. • Payment method is available.
Main Flow	1. Customer selects the payment option. 2. Customer enters credit card details. 3. System validates payment information. 4. System processes the payment. 5. System generates an invoice. 6. System sends the invoice to the customer via email.
Postconditions	Payment is completed and transaction is recorded.

Figure 2.4 : Payment Management Use Case

E5 – Delivery Management

Use Case	Update Delivery Status
Primary Actor	Delivery Personnel
Preconditions	• Delivery record exists for the order.
Main Flow	1. Delivery personnel logs into the system. 2. Delivery personnel selects the assigned delivery. 3. Delivery personnel updates the delivery status (shipped / in transit / delivered). 4. System records the updated status. 5. System notifies the customer about delivery progress.
Postconditions	Delivery status is updated and visible to the customer.

Figure 2.5 : Delivery Management Use Case

E6 – Feedback & Support

Use Case	Create Support Ticket
Primary Actor	Customer
Preconditions	• Customer is logged into the system.
Main Flow	1. Customer opens the support section. 2. Customer selects **Create Support Ticket**. 3. Customer enters issue details. 4. Customer submits the ticket. 5. System records the ticket. 6. System notifies the admin.
Postconditions	Support ticket is successfully created and available for admin review.

Figure 2.6 : Feedback and Support Management Use Case

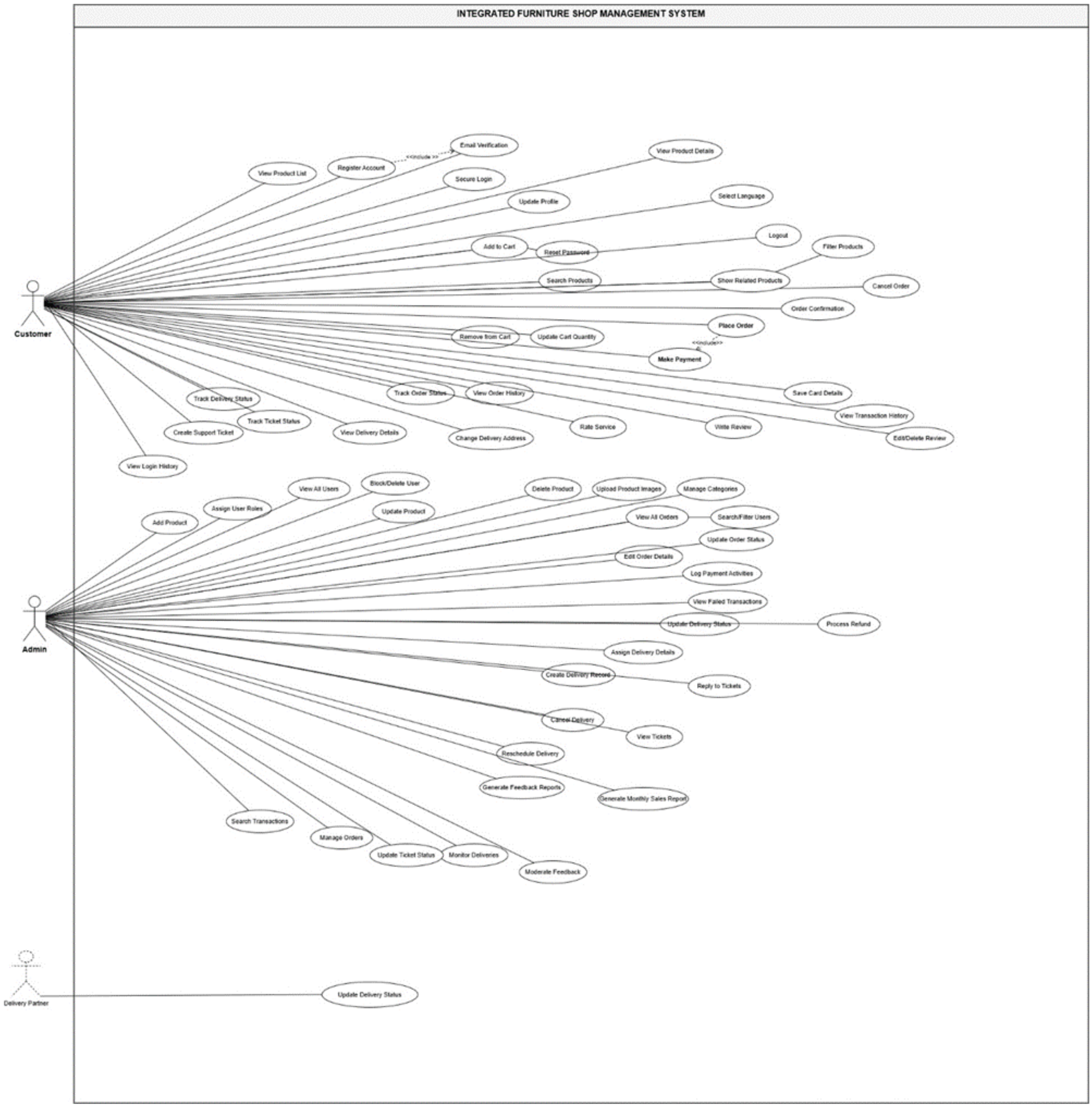

Figure 2.7: UML use case diagram

Further details: [UML use case diagram.pdf]

Activity diagrams shown in Figures 2.9 to 2.12 were developed to represent major operational workflows such as user registration, order processing, payment handling, and delivery coordination processes. These diagrams helped clarify system behavior and ensured consistency between stakeholder expectations and identified functional requirements. The corresponding diagrams are included in Appendix A and support the structured representation of system requirements identified during the analysis phase.

- User registration

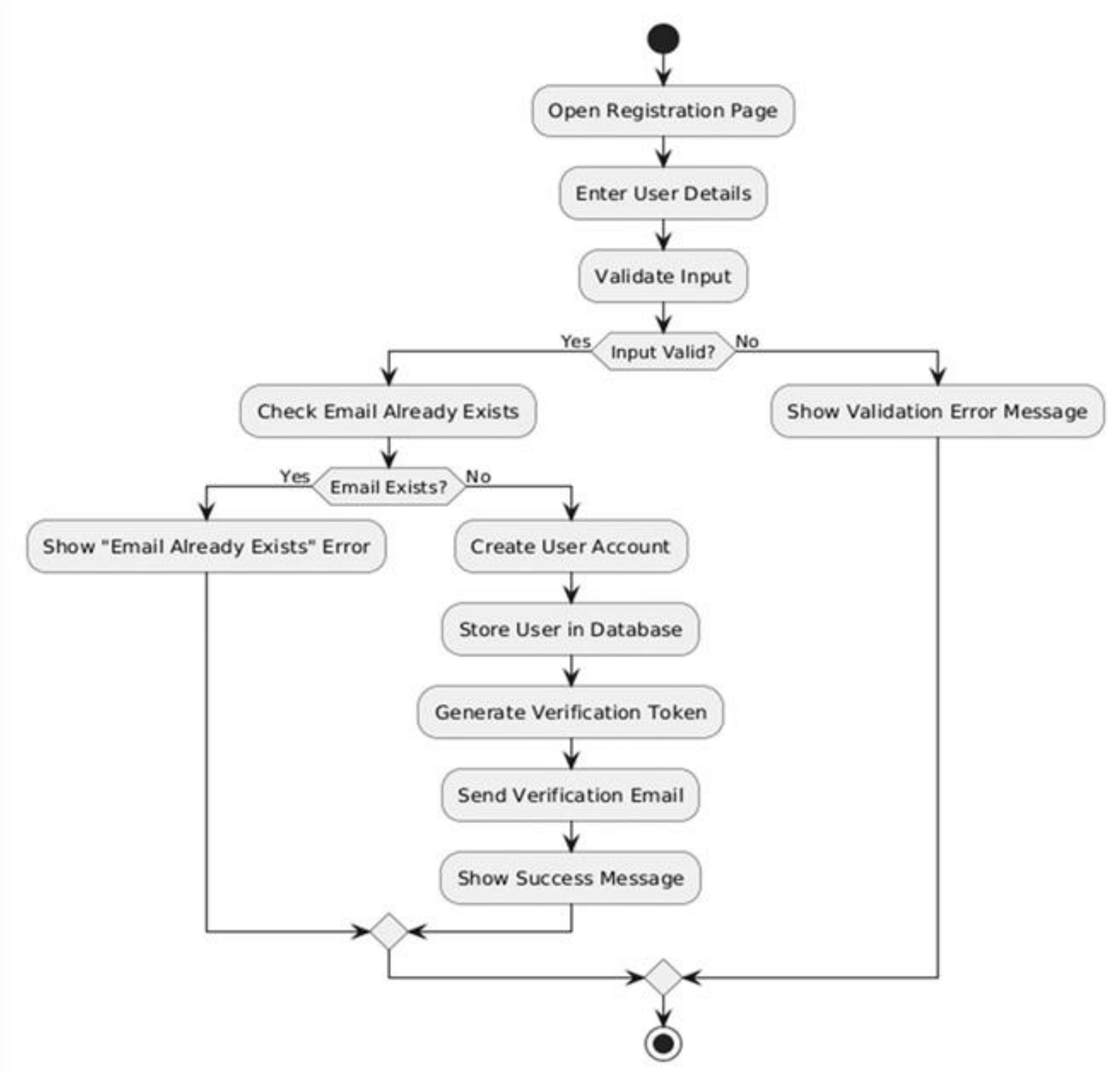

Figure 2.9 : User Management Activity Diagram

- Payment management

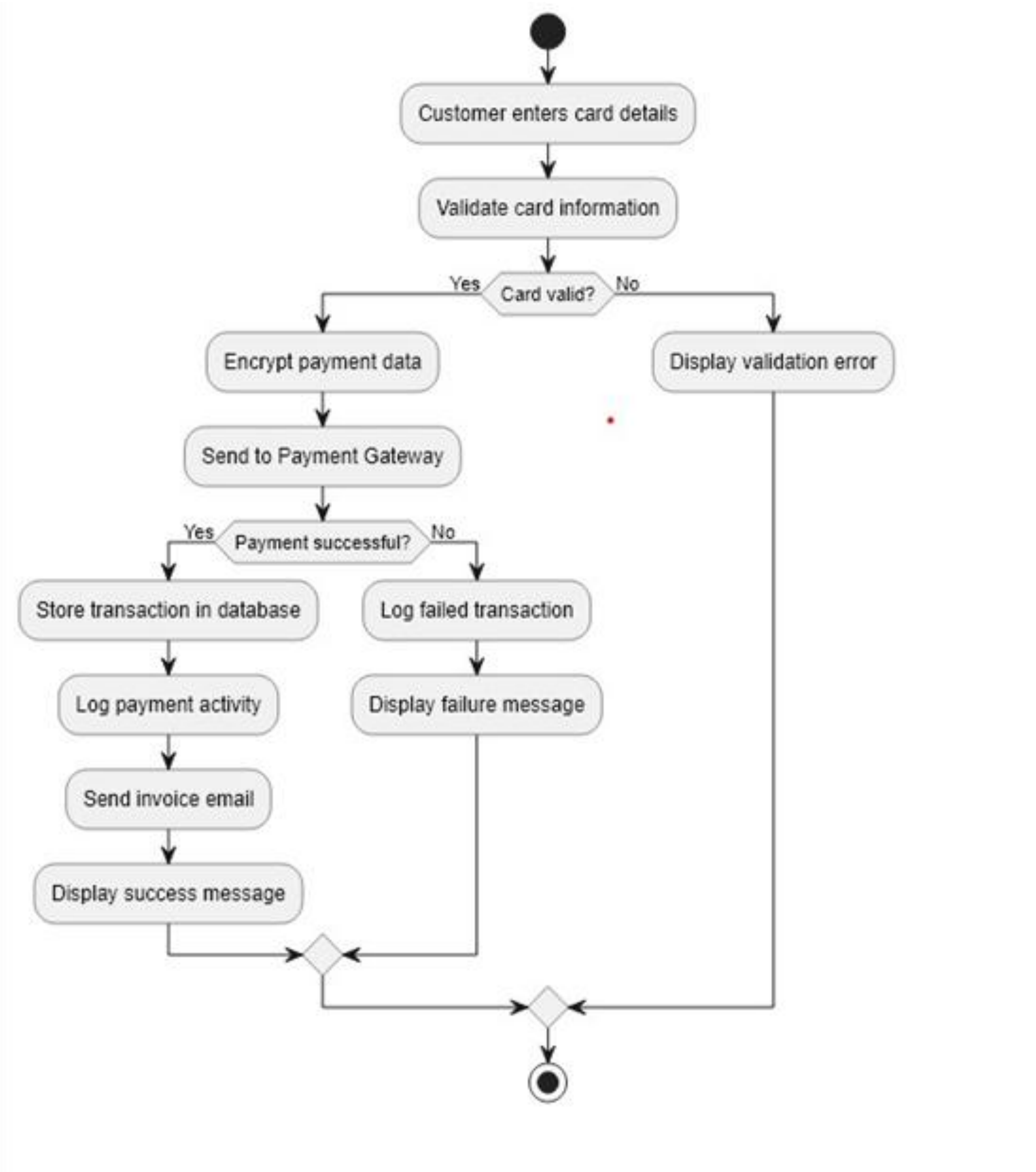

Figure 2.10 : Payment Management Activity Diagram

- Order Management

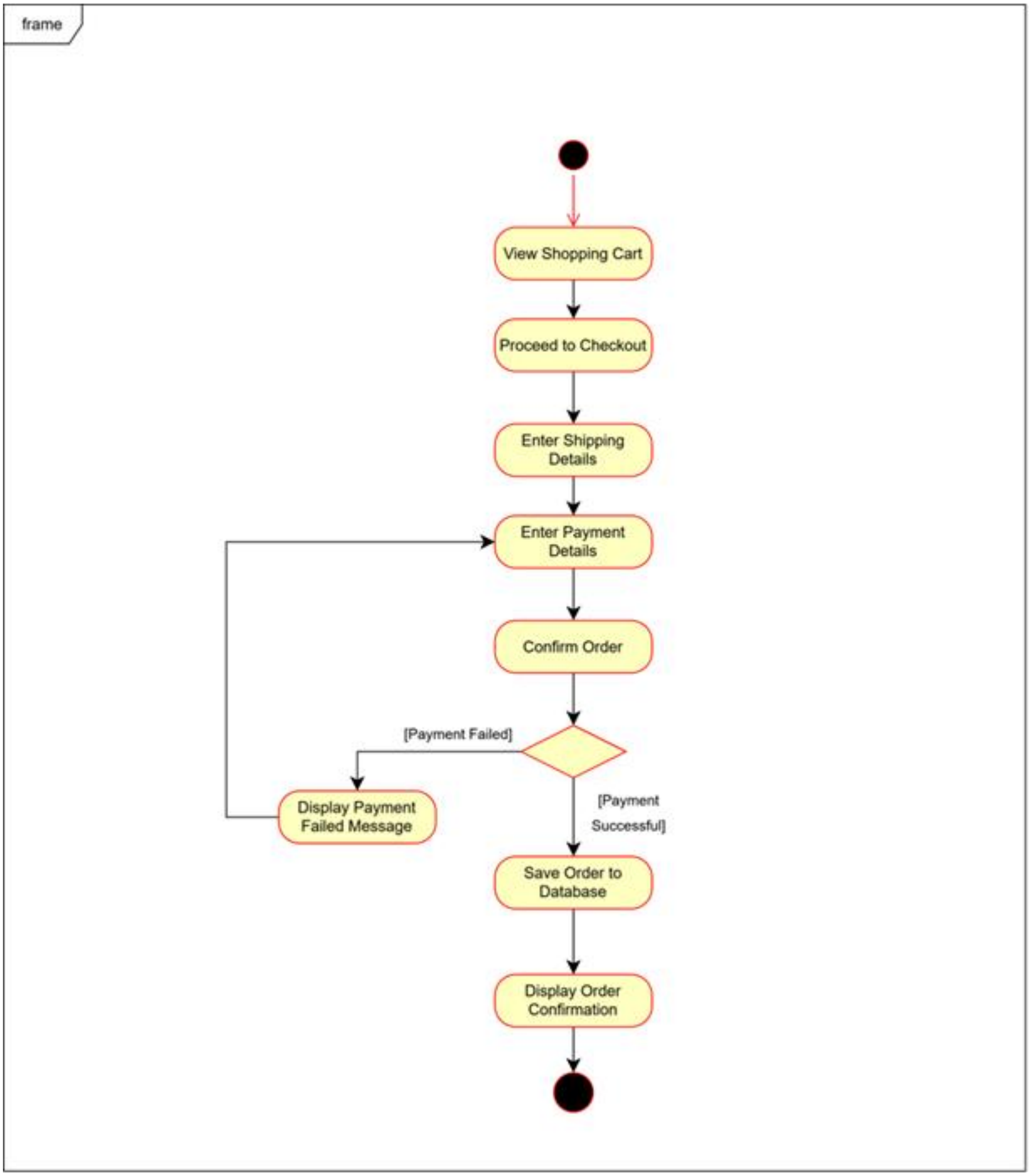

Figure 2.11 : Order Management Activity Diagram

- Delivery Management

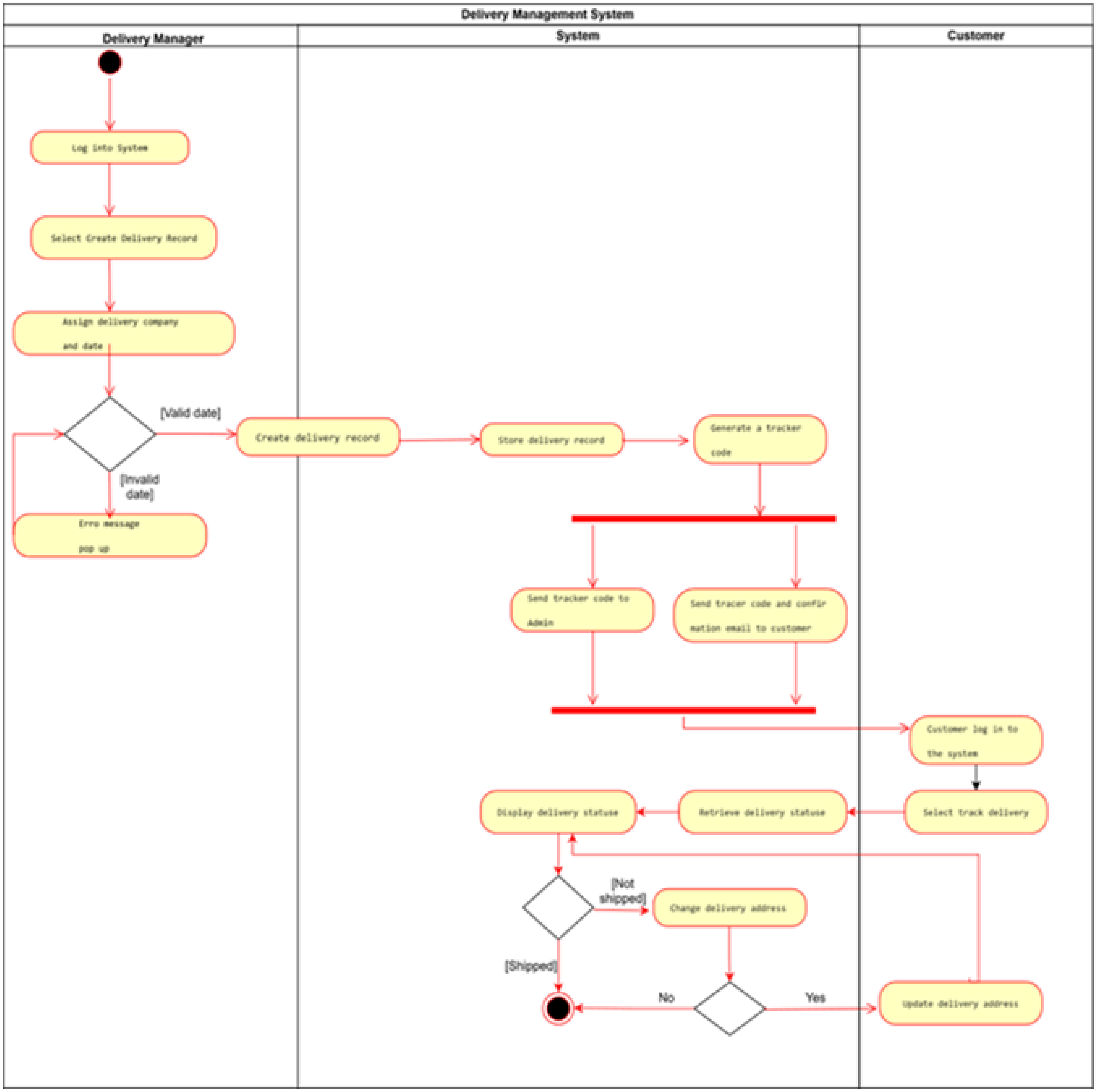

Figure 2.12 : Delivery Management Activity Diagram

2.2 Design

2.2.1 System Architecture Overview

The Integrated Furniture Shop Management System follows a centralized web-based client–server architecture that enables secure interaction between users and system services through a browser-based interface. The architecture separates the user interface, application logic, and database layers to ensure maintainability, scalability, and reliable system performance.

Customers, administrators, delivery personnel, and support staff access system functionality through client devices connected to the application server via secure communication protocols. The application server manages core operations such as product management, order processing, payment handling, delivery coordination, and customer support services, while a dedicated database server stores user records, product information, transaction details, delivery schedules, and support tickets.

External integrations such as payment gateway services and email notification systems support secure transaction processing and automated communication. This layered architecture ensures efficient workflow management and secure handling of system data across all modules.

2.2.2 UML Class Diagram

The UML Class Diagram (Figure 2.13)represents the structural design of the system by illustrating the main entities and their relationships within the furniture shop management environment. The primary classes identified include User, Product, Order, Payment, Delivery, and Support Ticket, which together support the core operational modules of the system.

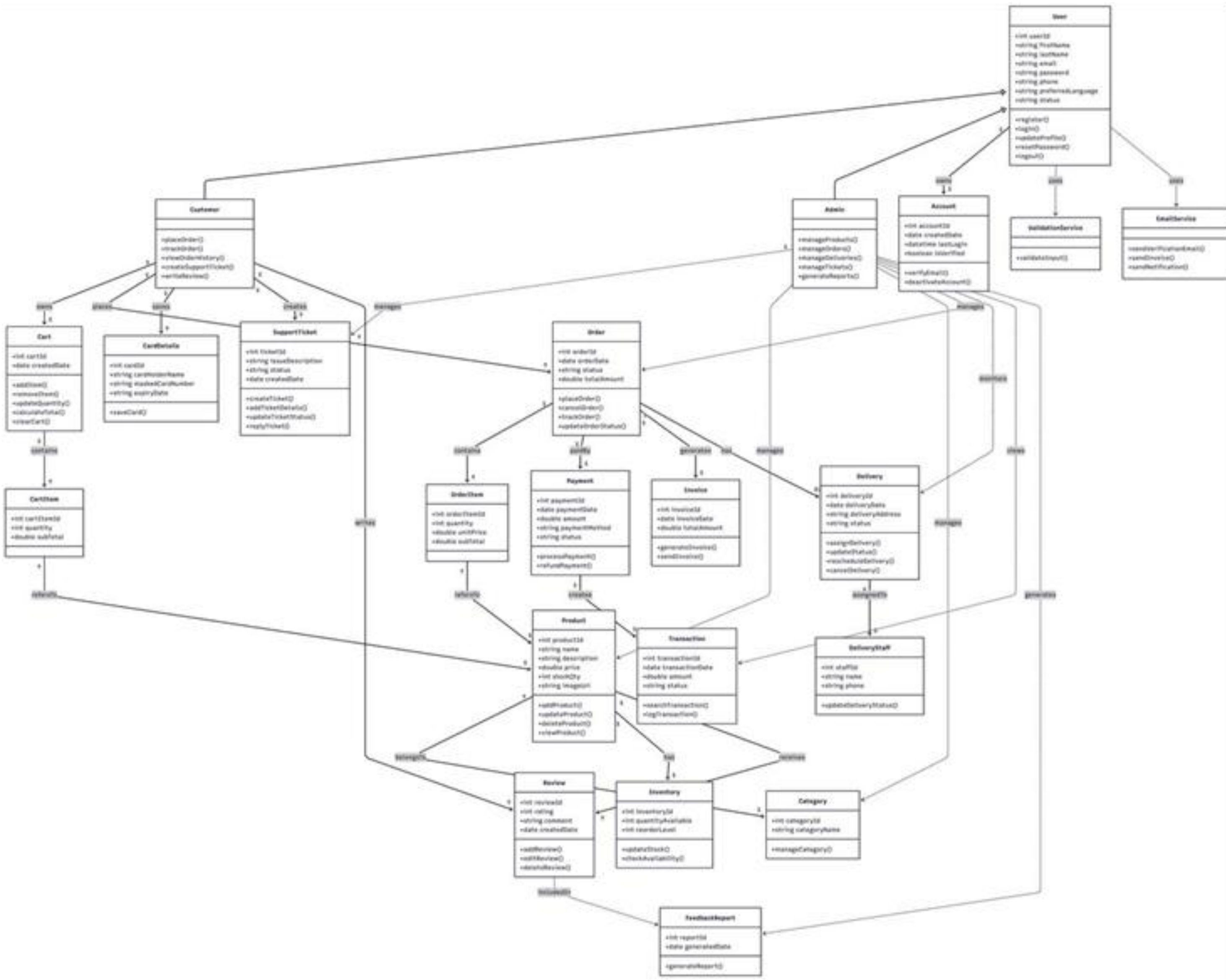

Figure 2.13 : UML Class Diagram

Further details : [UML class diagram.jpg]

2.2.3 Deployment Diagram

The Deployment Diagram(Figure 2.14) illustrates the physical architecture of the system and the distribution of software components across hardware nodes involved in system operation. Users access the system through client devices using web browsers connected via secure HTTPS communication to the application server.

The application server hosts the main system services responsible for executing business logic including order processing, product management, delivery coordination, and payment handling. A dedicated database server stores persistent system data such as user records, product information, transaction details, and delivery tracking information.

External services such as authentication providers and email notification servers are integrated to support secure login mechanisms and automated communication processes including account verification and invoice delivery. The deployment architecture ensures secure communication, efficient resource management, and reliable interaction between system components.

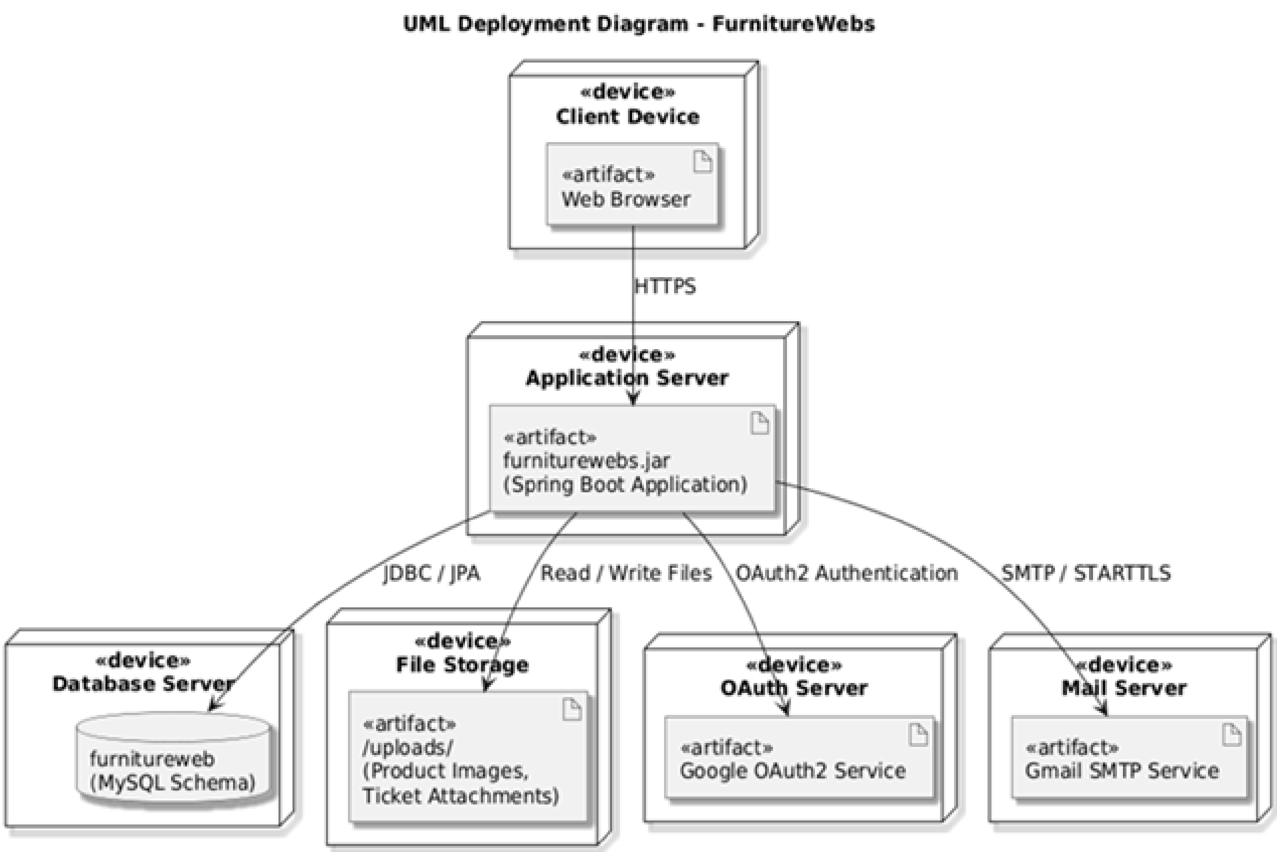

Figure 2.14 : Deployment Diagram

2.2.4 Sequence Diagrams

Sequence diagrams (Figure 2.15 to 2.18) were developed to illustrate interaction workflows between system components during major operational processes including user registration, order placement, payment processing, and delivery coordination activities.

- User Registration

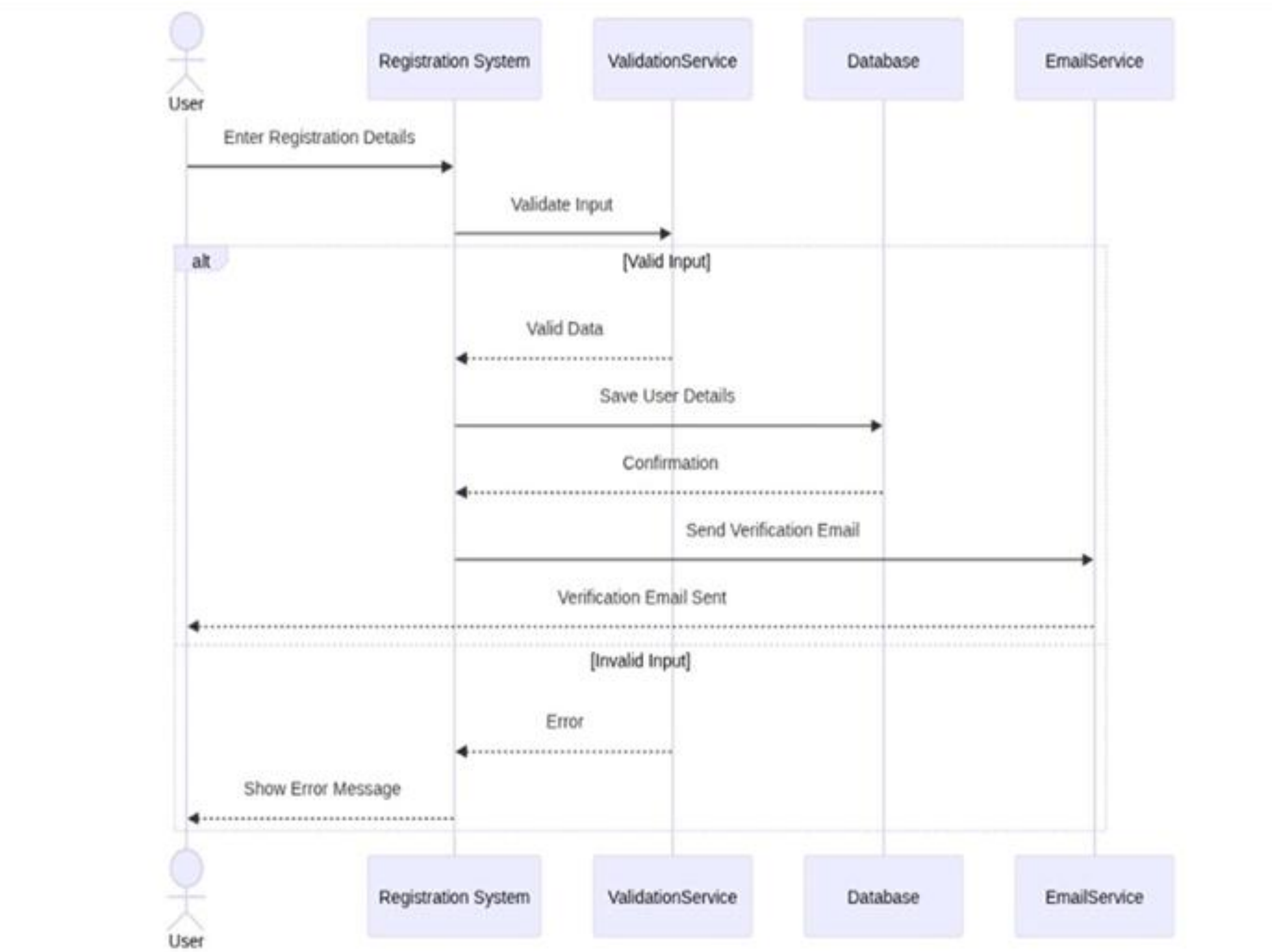

Figure 2.15 : User Management Sequence Diagram

- Order Management

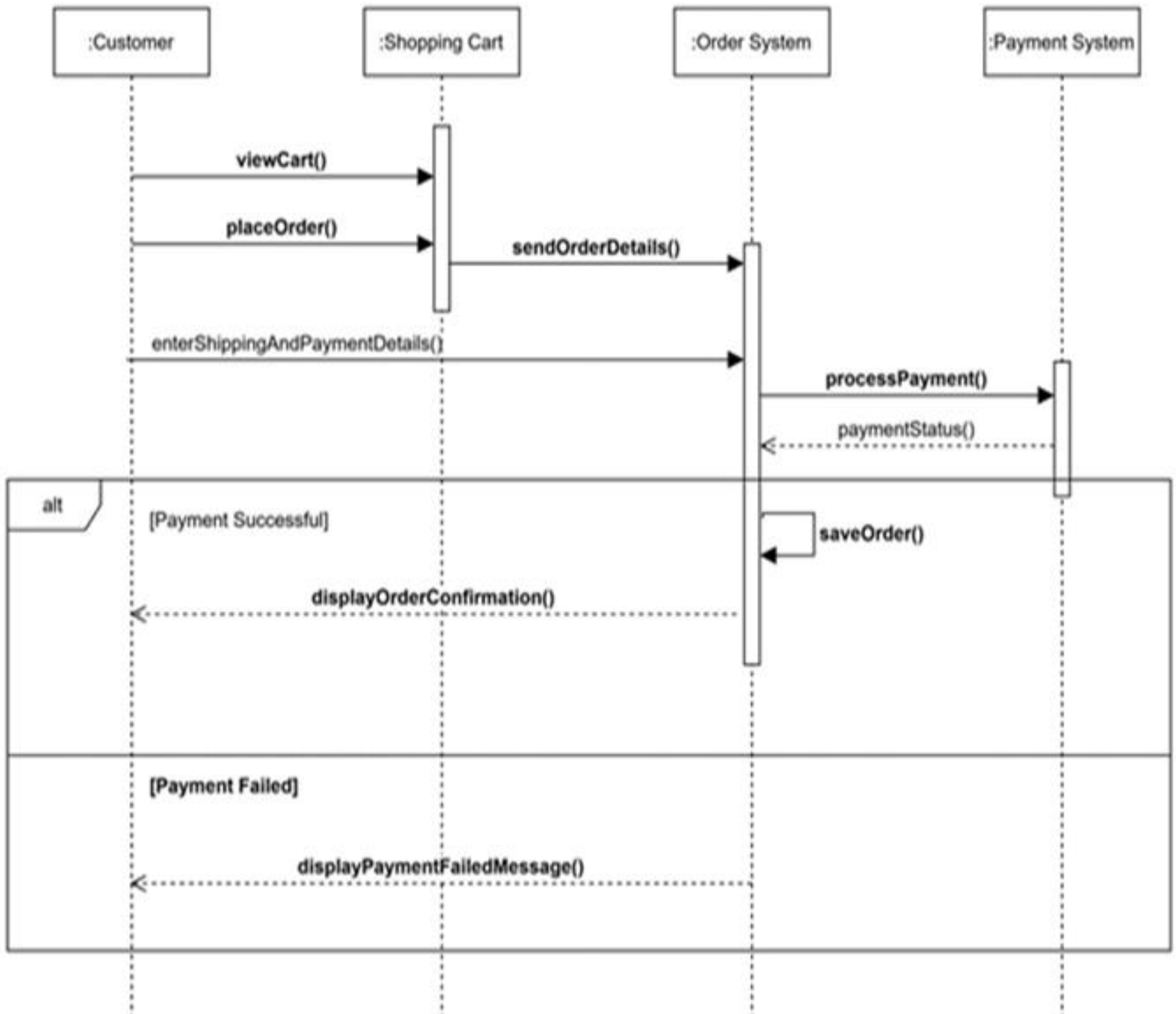

Figure 2.16 : Order Management Sequence Diagram

- Payment Management

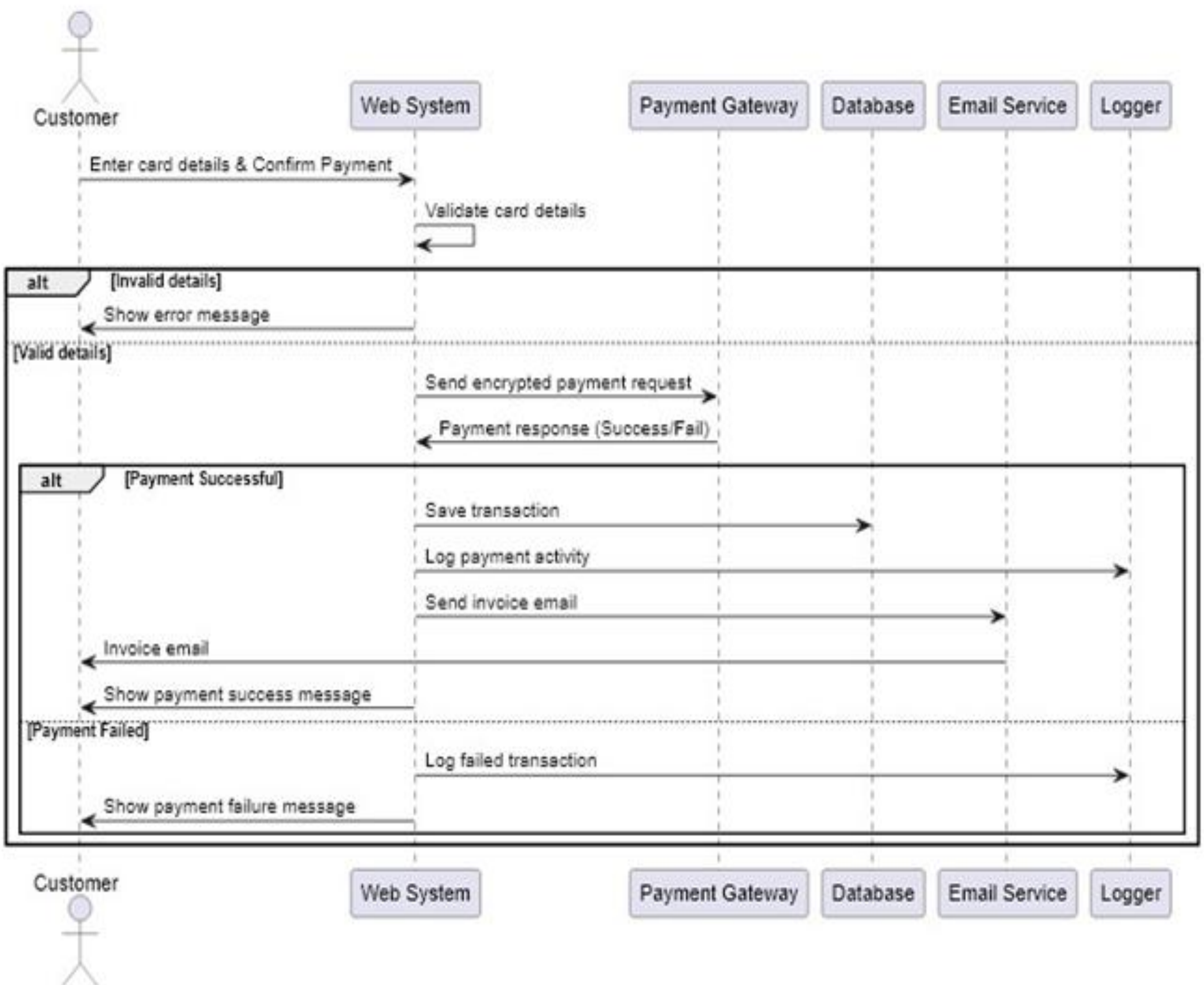

Figure 2.17 : Payment Management Sequence Diagram

- Delivery Management

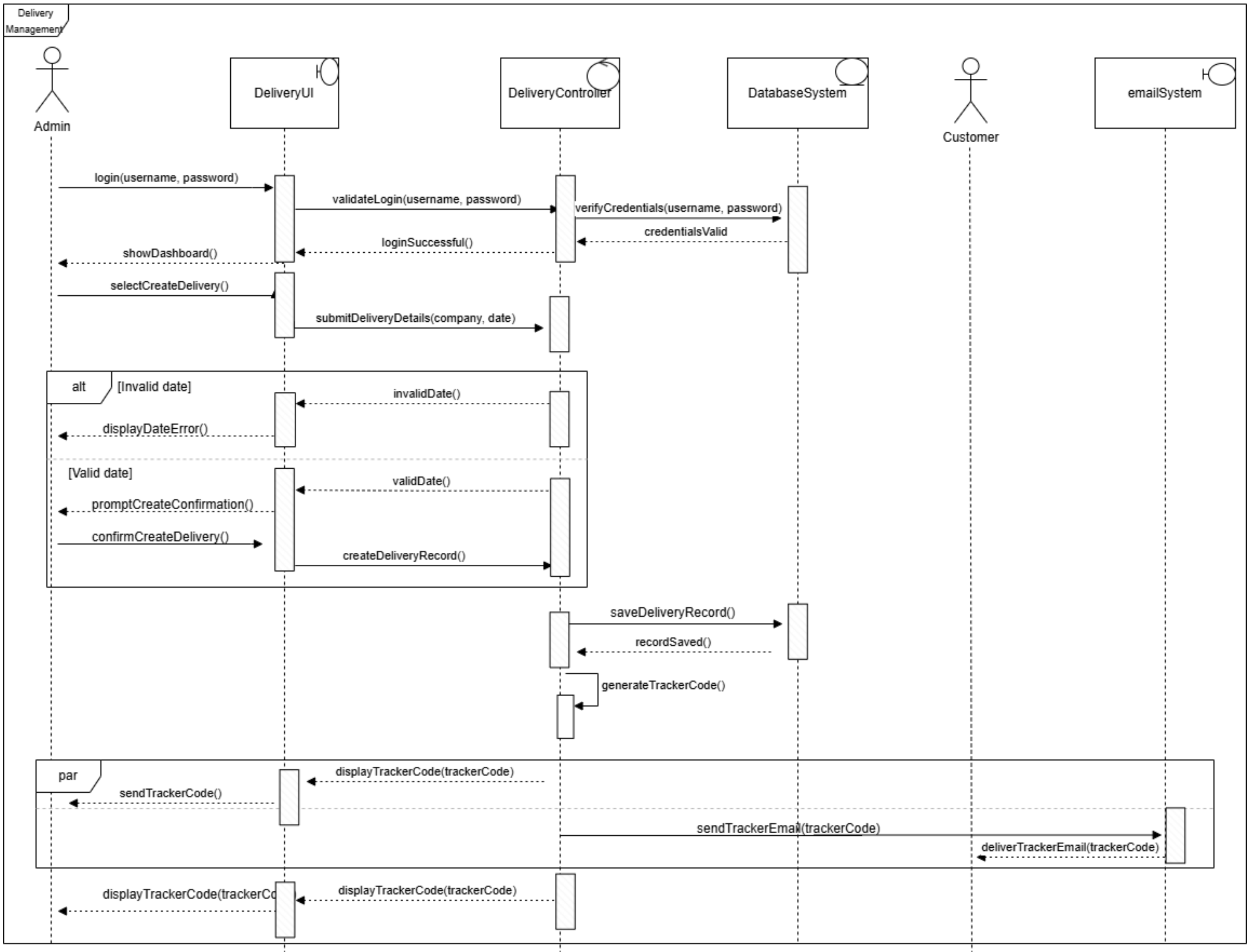

Figure 2.18 : Delivery Management Sequence Diagram

2.2.5 State Diagram

The State Diagram(Figure 2.19) represents the lifecycle transitions of customer orders within the Integrated Furniture Shop Management System. It illustrates how an order progresses through multiple states including pending, processing, shipped, in transit, out for delivery, and delivered.

The diagram also represents alternative transitions such as order cancellation before shipment dispatch. This structured representation ensures accurate tracking of order progress and supports coordination between order management and delivery tracking modules.

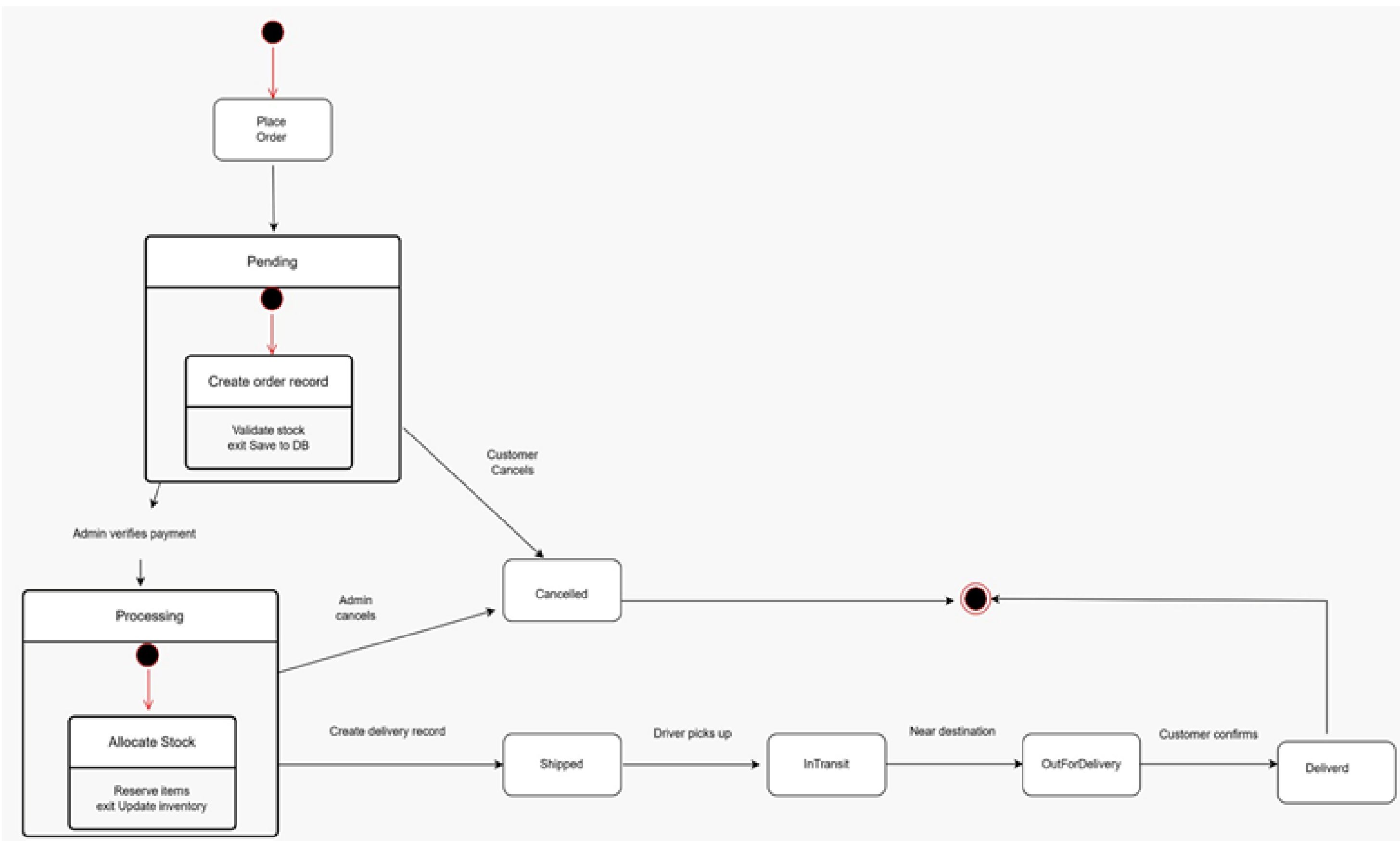

Figure 2.19 : UML State Diagram

2.2.6 Database Design (EER Model to Logical Schema)

The database design of the Integrated Furniture Shop Management System was developed using an Enhanced Entity–Relationship (EER) model (figure 2.20) representing core entities such as users, products, orders, payments, deliveries, and support tickets along with their relationships within the system. The EER model captures structural relationships between system components and supports accurate representation of operational workflows across multiple modules.

The conceptual EER model was transformed into a logical relational database schema (Figure 2.21) defining tables with primary and foreign key relationships to ensure data integrity and efficient interaction between system modules within the selected database management system environment.

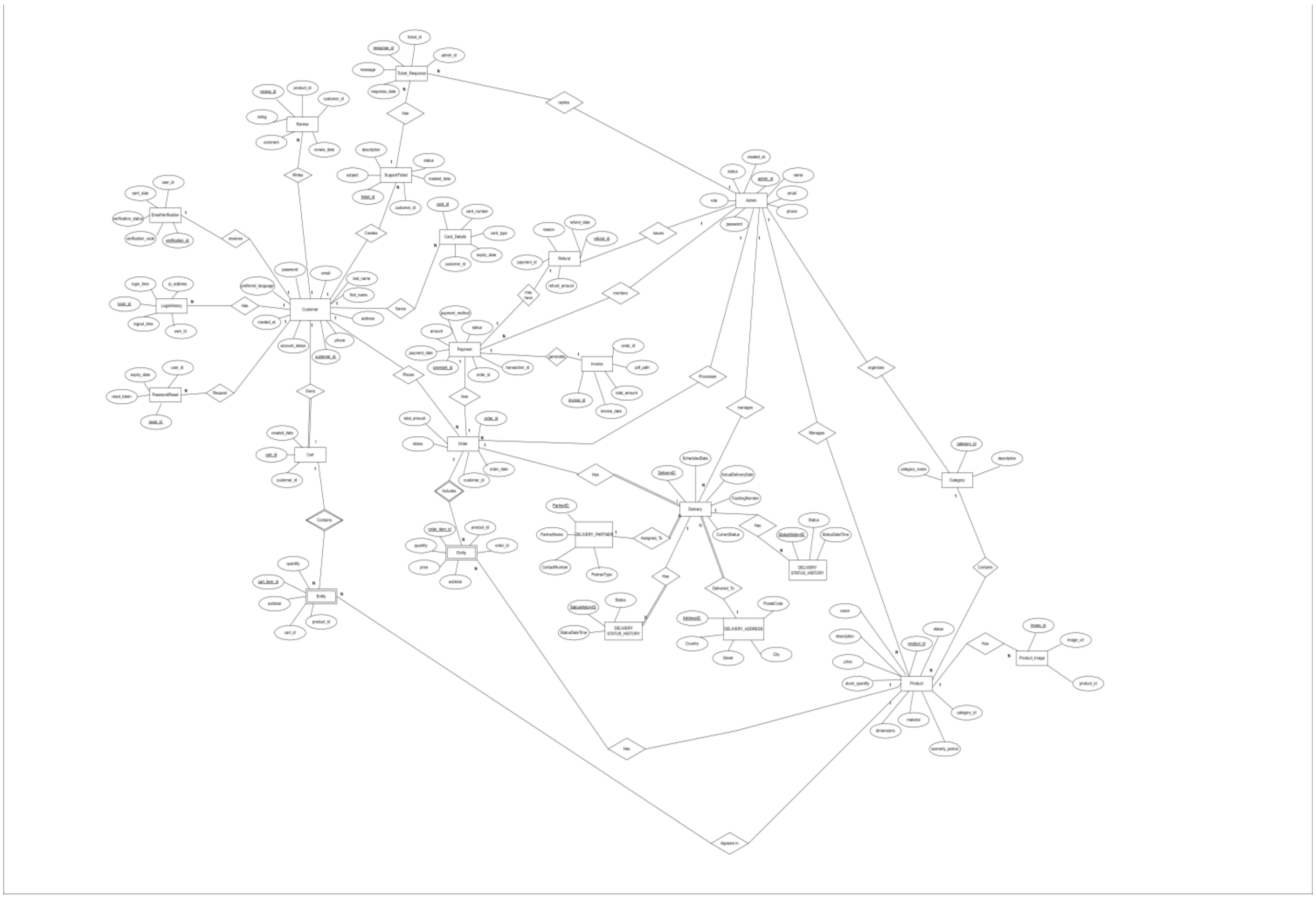

Figure 2.20 : EER Diagram

Further details : [EER Diagram.svg]()

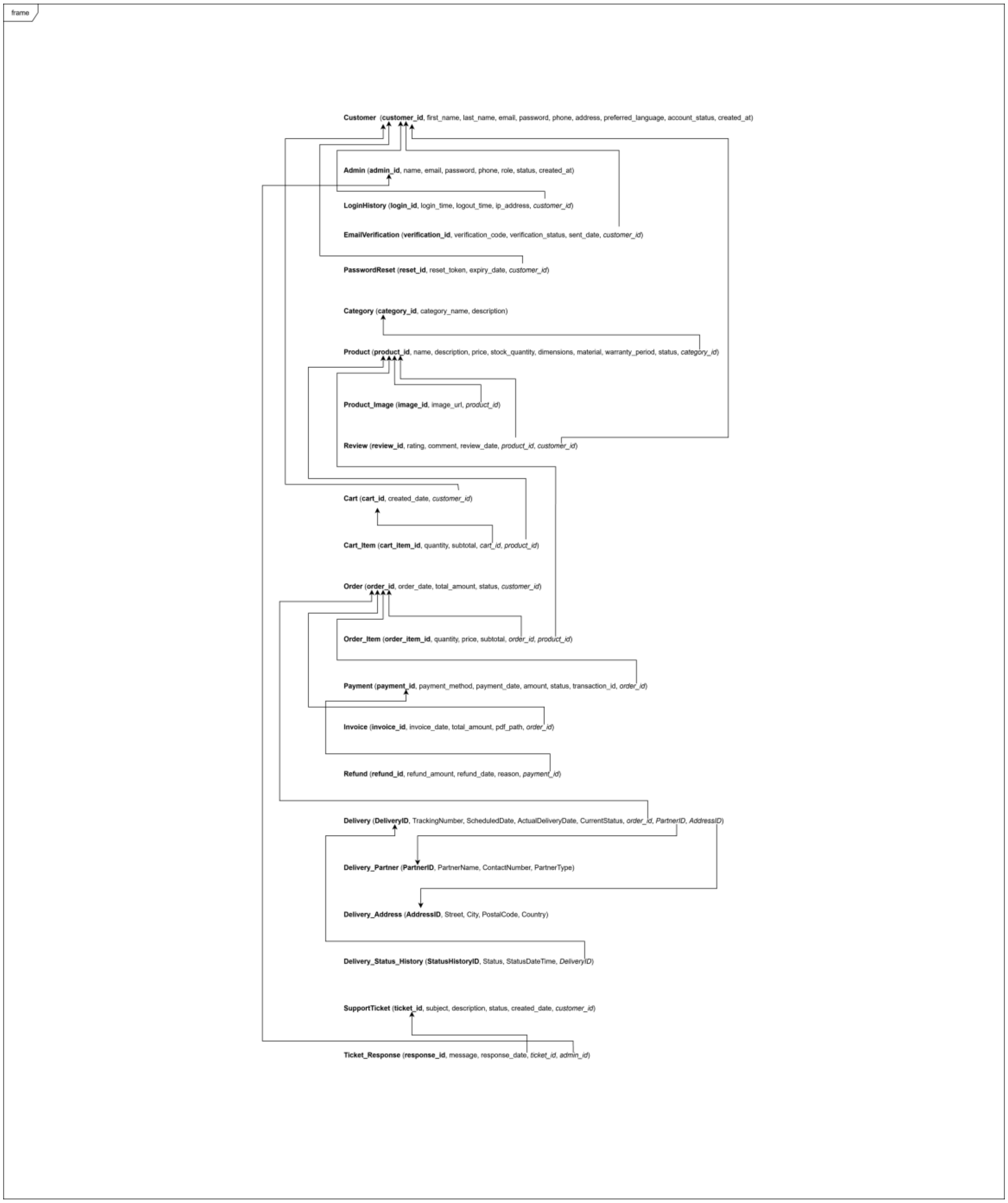

Figure 2.21: Relational Schema

Further details : [Relational Schema.svg]()

2.3 Implementation

2.3.1 Implementation Approach

The Integrated Furniture Shop Management System was implemented using a modular development approach based on the functional requirements identified during the requirements engineering phase. The system follows a layered web-based client–server architecture that separates presentation logic, application processing logic, and database services to improve maintainability and scalability.

The implementation process was carried out incrementally following Agile development practices, allowing individual modules such as user management, product management, order processing, payment handling, delivery coordination, and customer support management to be developed and tested progressively.

2.3.2 Development Tools and Technologies

The system was implemented using Java with the Spring Boot framework for backend development, HTML for frontend interface design, and MySQL as the relational database management system.

Spring Boot was selected because it simplifies backend service development through dependency management, built-in configuration support, and REST-based architecture capabilities. It also enables efficient integration between application logic and database services while supporting secure authentication workflows.

HTML was used to design structured user interface components that allow customers and administrators to interact with system services through a browser-based environment. MySQL was selected as the database management system due to its reliability, structured relational model support, and compatibility with Spring Boot through JDBC-based connectivity.

2.3.3 Module-Level Implementation

The system was implemented as six functional modules corresponding to the requirements identified during the analysis phase.

User and Account Management Module

This module was implemented using Spring Boot authentication mechanisms to support user registration, login validation, password recovery, and profile update functionalities. Role-based access control mechanisms were integrated to manage administrator and customer interactions within the system.

Product Management Module

The product management module enables administrators to create, update, and manage furniture product records including pricing details, descriptions, and category classification. This module supports real-time inventory visibility and allows customers to browse available products efficiently.

Shopping Cart and Order Management Module

This module enables customers to manage selected furniture items within a shopping cart interface and confirm purchase orders through structured order placement workflows. Order records are stored within the MySQL database and administrators can monitor order lifecycle progress through the system interface.

Payment and Billing Management Module

The payment module supports secure transaction handling by validating payment information and recording transaction details within the database. Automated invoice generation functionality ensures structured financial record maintenance within the system environment.

Delivery Management Module

The delivery management module enables administrators to schedule deliveries and assign shipment responsibilities while allowing customers to track delivery progress through status updates stored in the database.

Feedback and Support Management Module

This module allows customers to submit service feedback and support requests through the system interface. Administrators can monitor submitted requests and update resolution progress to maintain service quality and customer satisfaction.

2.3.4 Database Management System Selection

MySQL was selected as the relational database management system to support structured storage of system data including user information, product records, order transactions, delivery schedules, and support ticket details. The database design follows the logical schema derived from the Enhanced Entity Relationship (EER) model and uses primary and foreign key relationships to maintain referential integrity across system modules.

The integration between Spring Boot and MySQL was implemented using JDBC connectivity to ensure reliable data communication between the application server and database services.

2.3.5 Reusable Components and Framework Support

Spring Boot provides several reusable components that improved development efficiency, including dependency injection mechanisms, REST controller support, database connectivity configuration, and validation services. These reusable features reduced development complexity and ensured consistent interaction between application modules.

Standard validation mechanisms were implemented to verify user inputs during registration, order placement, and payment processing activities to improve system reliability and data accuracy.

2.4 Testing

Testing activities were conducted to ensure that all functional modules of the Integrated Furniture Shop Management System operated according to the specified requirements. The testing process focused on verifying system correctness, validating user interactions, and ensuring reliable communication between system components.

2.4.1 Test Levels

Testing was performed at multiple levels to validate system functionality and integration between modules.

Unit Testing : Unit testing was carried out to verify individual components such as authentication validation, product record updates, and order processing logic.

Integration Testing :Integration testing ensured correct interaction between modules including database connectivity, order-payment communication, and delivery status updates.

System Testing :System testing validated the complete workflow of the application including user registration, order placement, payment confirmation, and delivery tracking processes within the deployed environment.

User Acceptance Testing :User acceptance testing confirmed that system functionality satisfied stakeholder expectations and supported operational workflows identified during the requirements analysis phase.

2.4.2 Test Cases

- User Management Test Cases

Table 2.4 : User Management Test Cases

Test Case ID	Test Case Description	Test Steps	Acceptance Criteria	Expected Result	Test Result
UA001	Register new user	1. Navigate to	User account	User account stored	Pass

		Registration page 2. Enter user details 3. Click "Register"	created successfully	in database	
UA002	*User login*	1. Navigate to Login page 2. Enter valid credentials 3. Click "Login"	Valid credentials allow access	User redirected to dashboard	Pass
UA003	Reset password	1. Click "Forgot Password" 2. Enter registered email 3. Submit request	Reset link sent successfully	Password reset email received	Pass
UA004	Update profile	1. Navigate to Profile page 2. Modify details 3. Click "Save"	Updated data stored	Profile updated successfully	Pass
UA005	Logout use	1. Click "Logout" button	User session terminated successfully	User redirected to login page	Pass

- Product Management Test Cases

Table 2.5: Product Management Test Cases

Test Case ID	**Test Case Description**	**Test Steps**	**Acceptance Criteria**	**Expected Result**	**Test Result**
PM001	Add product	1. Navigate to Product Management 2. Click "Add Product"	Product is successfully added to inventory	Product record saved	Pass

		3. Enter product details 4. Click "Save"			
PM002	View product list	1. Navigate to Product Management 2. Click "View Products"	Product list is displayed	All products are shown	Pass
PM003	Update product details	1. Select product 2. Click "Edit" 3. Modify details 4. Click "Save"	Updated product details stored	Product details updated successfully	Pass
PM004	Delete product	1. Select product 2. Click "Delete" 3. Confirm deletion	Product removed from system	Product record deleted	Pass
PM005	Search product	1. Enter product name in search bar 2. Click "Search"	Matching product displayed	Correct product shown in results	Pass

- Shopping cart and Order Management Test Cases

Table 2.6: Shopping cart and Order Management Test Cases

Test Case ID	**Test Case Description**	**Test Steps**	**Acceptance Criteria**	**Expected Result**	**Test Result**
OM001	Add product to cart	1. Select product 2. Click "Add to Cart"	Product added to cart	Cart updated successfully	Pass
OM002	Update cart quantity	1. Navigate to Cart 2. Modify item quantity 3. Click "Update"	Cart updated with new quantity	Quantity updated correctly	Pass

OM003	Place order	1. Navigate to Checkout 2. Confirm order details 3. Click "Place Order"	Order recorded successfully	Order stored in database	Pass
OM004	View order history	1. Navigate to Order History	Order history displayed	Orders listed correctly	Pass
OM005	Cancel order	1. Navigate to Order History page 2. Select pending order 3. Click **Cancel Order** button 4. Confirm cancellation	Selected order is cancelled successfully before processing stage	Order status updated to **Cancelled** and removed from active orders list	Pass

- **Payment Module Test Cases**

Table 2.7: Payment Module Test Cases

Test Case ID	**Test Case Description**	**Test Steps**	**Acceptance Criteria**	**Expected Result**	**Test Result**
PAY001	Process payment	1. Navigate to Payment page 2. Enter payment details 3. Click "Confirm Payment"	Payment processed successfully	Payment confirmation displayed	Pass
PAY002	Generate invoice	1. Complete payment process	Invoice generated automatically	Invoice available for viewing	Pass
PAY003	View transaction history	1. Navigate to Transaction	Transaction list displayed	Payment records shown correctly	Pass

		History page			
PAY004	Validate payment details	1. Enter invalid payment details 2. Submit payment	Invalid details rejected	Error message displayed	Pass
PAY005	Confirm payment status update	1. Complete payment process 2. Check order status	Payment status updated	Order marked as paid	Pass

- Delivery Module Test Cases

Table 2.8: Delivery Module Test Cases

Test Case ID	**Test Case Description**	**Test Steps**	**Acceptance Criteria**	**Expected Result**	**Test Result**
DL001	Assign delivery	1. Navigate to Delivery Management 2. Assign delivery manager. 3. Save assignment	Delivery assigned successfully	Delivery record stored	Pass
DL002	Update delivery status	1. Select delivery record 2. Update status 3. Save changes	Delivery status updated	Status change reflected correctly	Pass
DL003	Track delivery progress	1. Navigate to Delivery Tracking page	Delivery progress displayed	Delivery status visible to user	Pass
DL004	Modify delivery address	1. Select delivery record 2. Edit address	Address updated successfully	New address saved	Pass

		3. Save changes			
DL005	View delivery schedule	1. Navigate to Delivery Schedule page	Delivery schedule displayed	Scheduled deliveries visible	Pass

- Support Module Test Cases

Table 2.9: Support Module Test Cases

Test Case ID	Test Case Description	Test Steps	Acceptance Criteria	Expected Result	Test Result
FS001	Submit support ticket	1. Navigate to Support page 2. Enter issue details 3. Click "Submit"	Support ticket created successfully	Ticket stored in system	Pass
FS002	View support response	1. Navigate to Support History page	Support response displayed	Response visible to user	Pass
FS003	Submit product review	1. Navigate to Product page 2. Enter review 3. Click "Submit"	Review stored successfully	Review displayed correctly	Pass
FS004	Submit product review	1. Navigate to Product page 2. Enter review 3. Click "Submit"	Review submitted successfully	Review displayed correctly	Pass
FS005	View customer feedback reports	1. Navigate to Feedback Reports page	Feedback summary displayed	Reports generated correctly	Pass

3. Evaluation

3.1 Assessment of the Project results

The results of the Integrated Furniture Shop Management System were evaluated by comparing the implemented system functionality with the objectives defined during the requirements engineering phase and by validating operational workflows through functional testing activities. System behavior was analyzed using module-level testing results, user interaction validation, and execution of key business workflows including registration, product management, order processing, payment handling, delivery tracking, and support request management.

Screenshots of the implemented system interfaces (Figure 3.1 to 3.6) are provided as evidence of successful module implementation and functional integration across the system environment.

3.1.1 Implementation Results

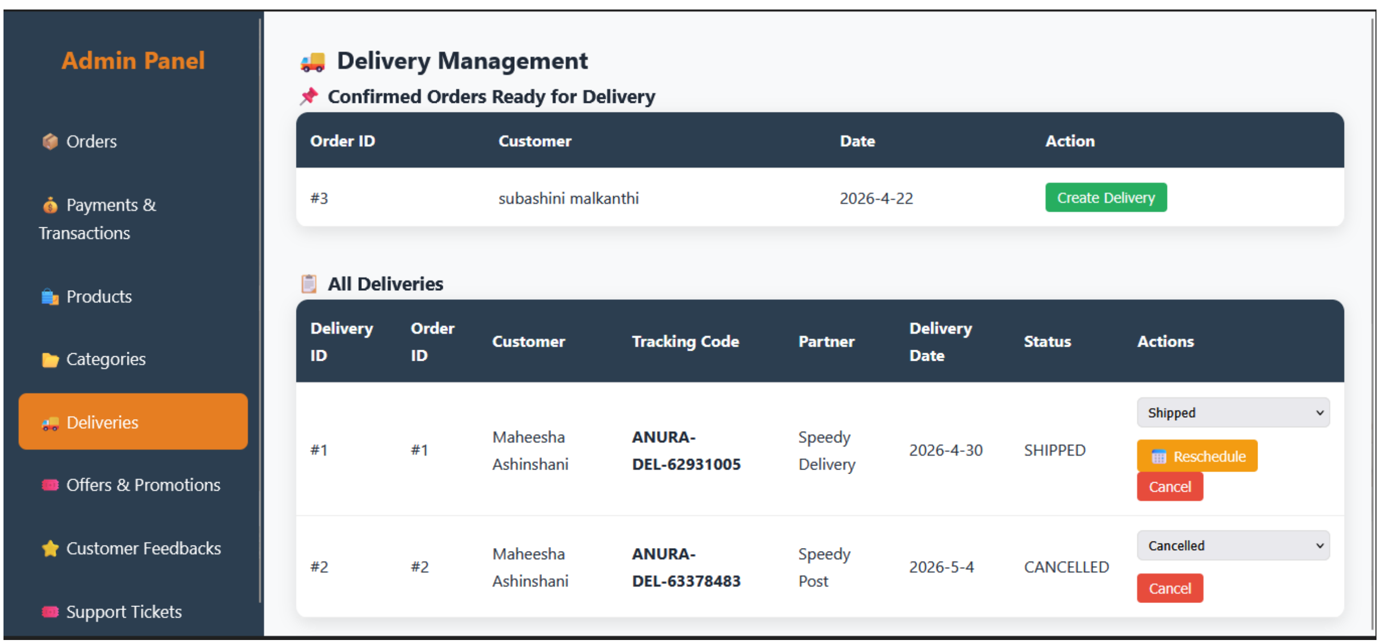

Figure 3.1: Delivery Management proof

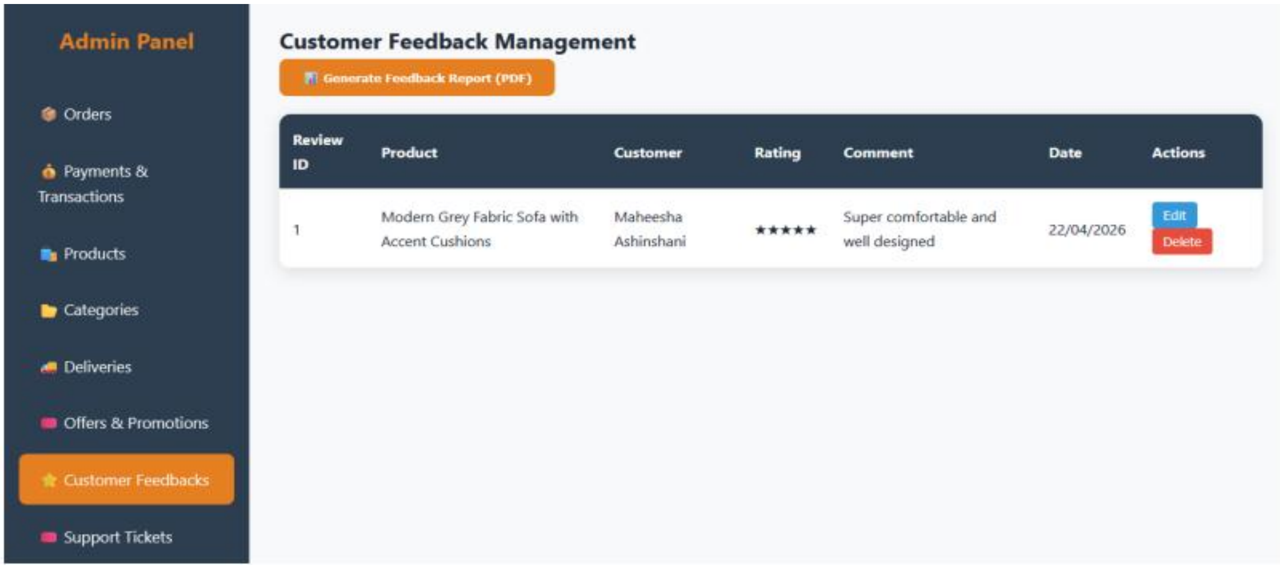

Figure 3.2 : Customer Feedback Management Proof

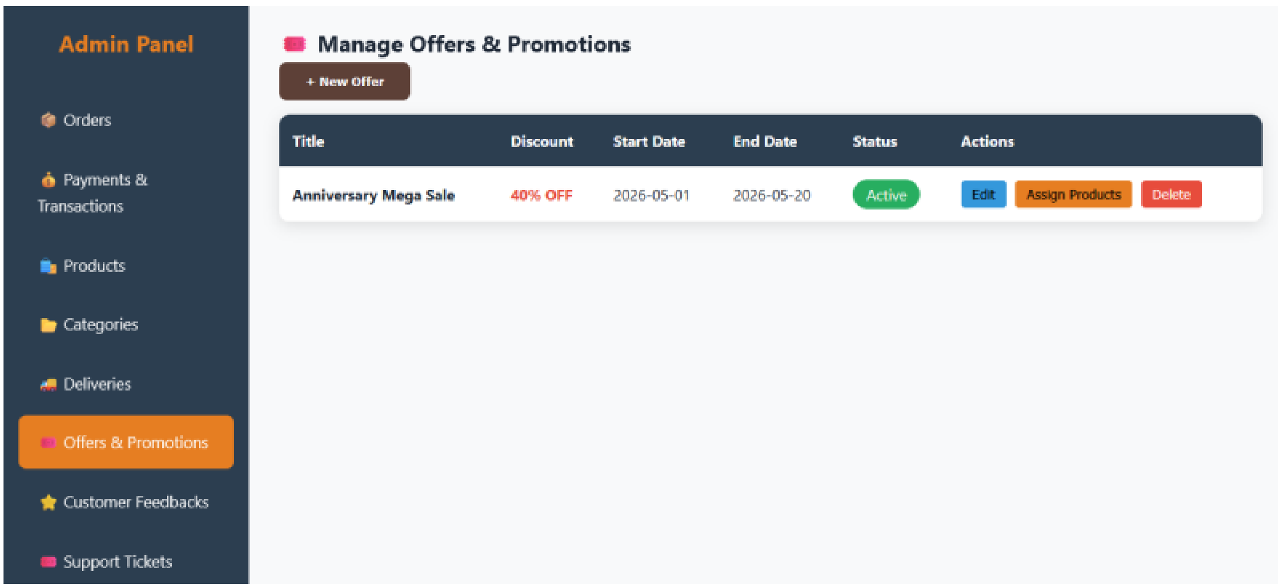

Figure 3.4 : Offers & Promotion proof

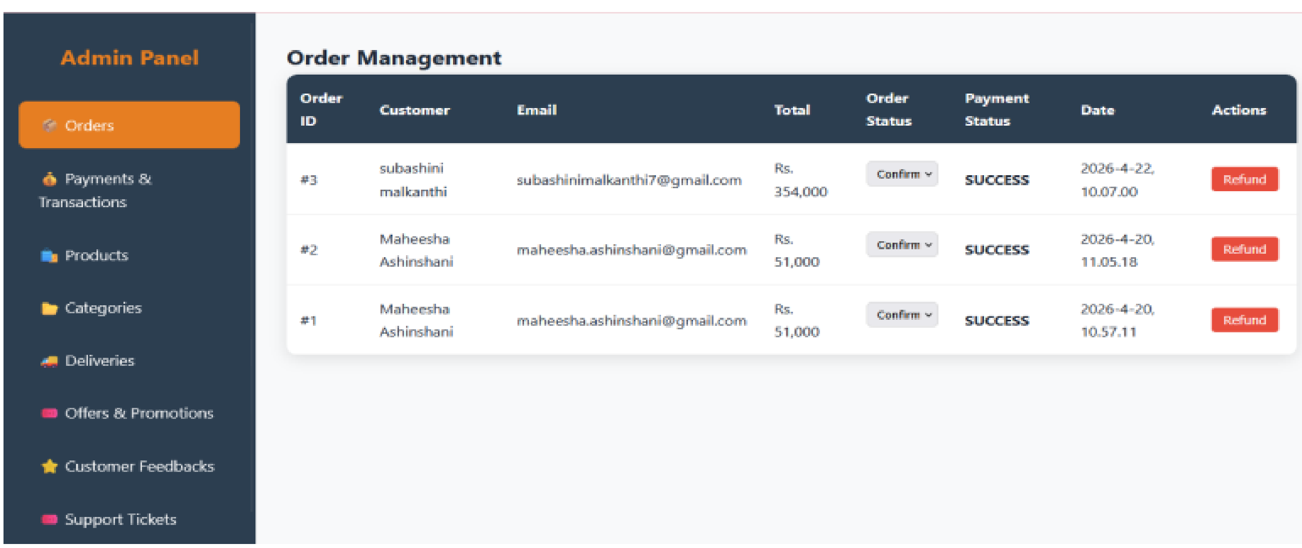

Figure 3.3 : Order Managment Proof

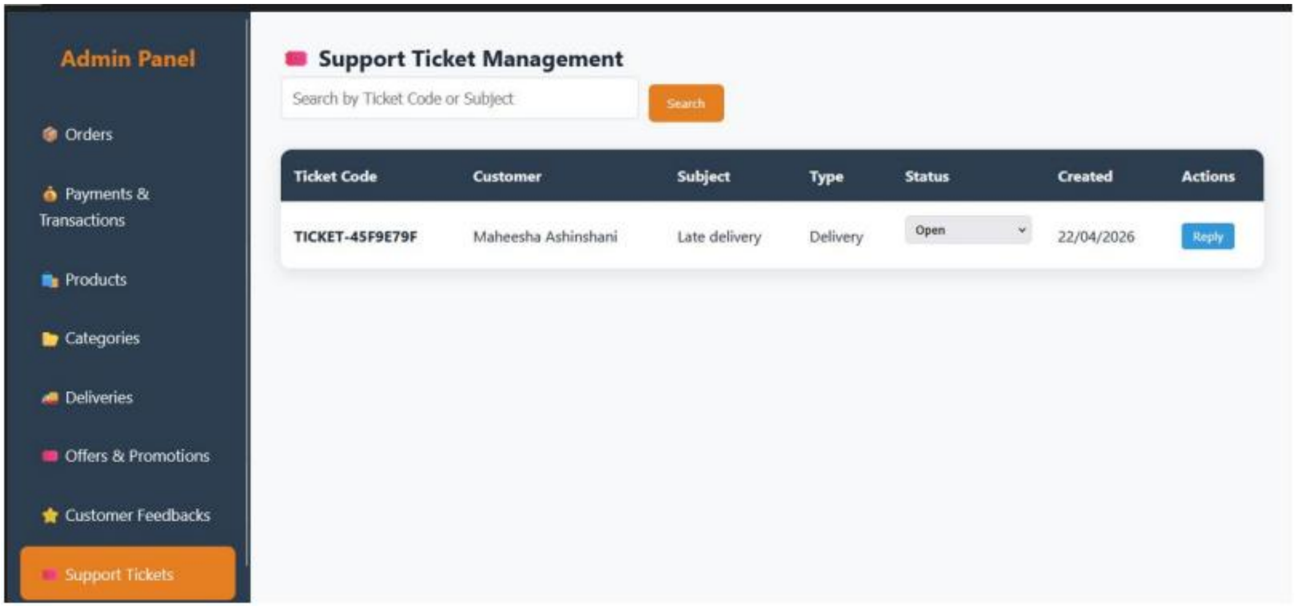

Figure 3.6 : Support Ticket Proof

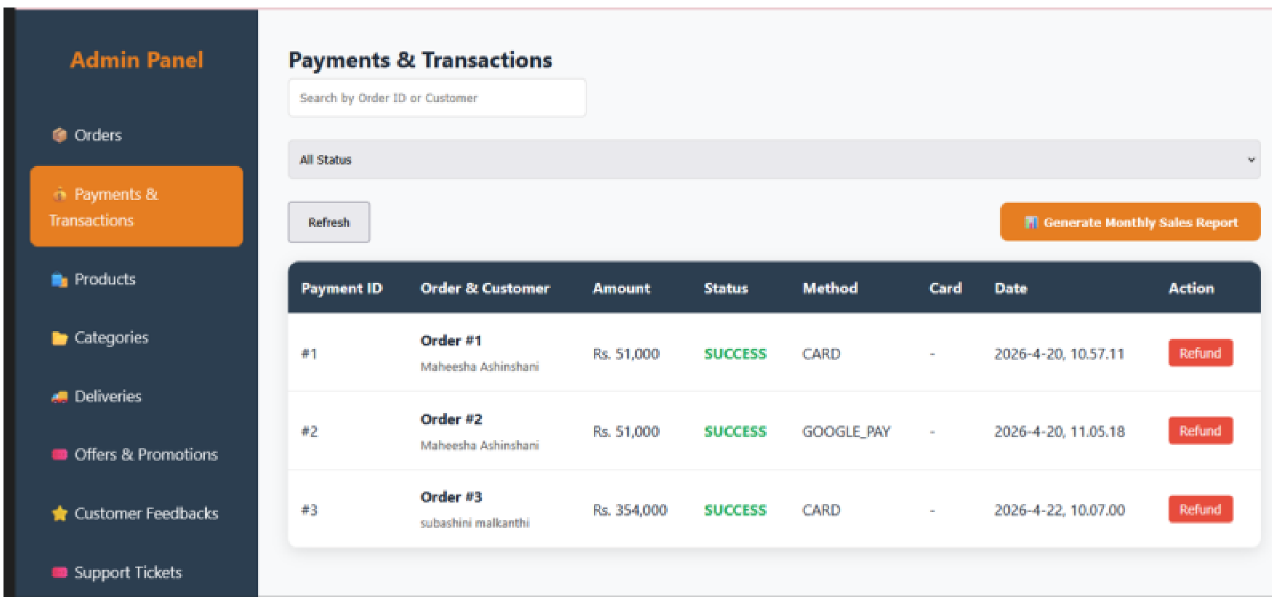

Figure 3.5 : Payment proof

3.1.2 Objective Achievement Analysis

The implemented system successfully achieved the major objectives defined during the project planning phase. The user management module supports secure registration and authentication workflows, while the product management module enables administrators to maintain accurate product records within the system database. The shopping cart and order management module allows customers to place and monitor orders efficiently, and the payment module supports structured transaction recording and invoice generation.

In addition, the delivery management module enables administrators to assign delivery schedules and update shipment status information, while the support management module allows customers to submit feedback and service requests through the system interface. These results confirm that the system satisfies the primary functional requirements identified during the requirements analysis phase.

3.1.3 Identified Limitations

Although the system successfully implements the core functional requirements, several limitations were identified during evaluation. Advanced analytics features such as sales trend visualization and automated reporting dashboards were not included in the current implementation. Integration with external supplier management systems was also outside the scope of the project.

In addition, mobile application support was not implemented in the current version of the system, and payment gateway integration was implemented at a functional level without full-scale deployment testing in a production environment.

3.2 Lessons Learned

The development of the Integrated Furniture Shop Management System provided several important technical and project management learning experiences throughout the requirements analysis, design, implementation, and testing phases of the project lifecycle.

One key lesson learned during the requirements engineering phase was the importance of using multiple elicitation techniques such as stakeholder interviews, workflow observation, and document analysis to accurately identify operational challenges within the existing furniture shop environment. These techniques helped ensure that system requirements were aligned with actual business needs and supported the identification of structured functional modules.

During the system design phase, the use of modelling techniques such as use case diagrams, activity diagrams, sequence diagrams, class diagrams, and Enhanced Entity Relationship (EER) diagrams helped improve understanding of system workflows and relationships between system components. These modelling approaches ensured consistency between requirements and implementation decisions and supported effective communication among development team members.

The implementation phase provided practical experience in developing a layered web-based application using Java and the Spring Boot framework. The integration of backend services with the MySQL database improved understanding of relational database design concepts such as primary and foreign key relationships, structured data storage, and database connectivity using JDBC. The use of modular development practices also demonstrated the importance of separating application logic from user interface components to improve system maintainability.

Another important lesson learned during the project was the significance of performing structured testing activities to validate system functionality. Executing module-level test cases helped identify potential input validation issues and ensured correct interaction between application modules such as order processing, payment handling, and delivery tracking workflows.

The project also highlighted the importance of effective time management and task coordination when working within an Agile development environment. Sprint-based planning activities helped organize development tasks and supported incremental implementation of system modules according to priority levels defined during the requirements engineering phase.

Overall, the project provided valuable experience in applying software engineering principles to develop a real-world business management system and demonstrated the importance of structured requirements analysis, system modelling, modular implementation, and testing practices in delivering reliable software solutions.

3.3 Future Work

The following enhancements can be considered for future versions of the Integrated Furniture Shop Management System to further improve functionality, usability, and scalability:

- Implement advanced reporting and analytics dashboards to visualize sales trends, inventory usage patterns, and customer purchasing behavior for improved business decision-making.
- Develop a mobile application version of the system to allow customers and administrators to access services such as order tracking and delivery updates through smartphones.
- Integrate a supplier management module to support automated stock replenishment and improve coordination between the furniture shop and suppliers.
- Introduce multi-factor authentication mechanisms to enhance system security and protect sensitive user and transaction data.
- Implement a personalized product recommendation feature based on customer browsing history and purchasing behavior to improve user experience.
- Integrate real-time notification services such as email or SMS alerts for order confirmations, payment updates, and delivery tracking notifications.
- Extend the system to support role-based reporting tools that provide administrators with detailed operational summaries and performance indicators.
- Improve scalability by deploying the system using cloud-based infrastructure services to support future expansion and increased user activity.

4. Conclusion

The Integrated Furniture Shop Management System was successfully developed to address operational challenges identified in the existing furniture retail environment by introducing a centralized web-based platform for managing inventory, orders, payments, deliveries, and customer support activities. The system achieved its primary objectives by improving workflow efficiency, reducing manual record-handling errors, and enabling structured transaction monitoring within the organization.

The implementation using Java, Spring Boot, HTML interfaces, and a MySQL relational database supported reliable interaction between system modules and ensured secure storage of operational data. System testing confirmed that all major functional modules operated according to the specified requirements and supported the intended business workflows.

Although the system successfully achieved its core functional objectives, some limitations such as the absence of advanced reporting dashboards, supplier integration features, and mobile accessibility were identified. These limitations can be addressed in future versions through analytics modules, supplier coordination functionality, and mobile-based system extensions.

Overall, the proposed system provides significant benefits to the organization by improving inventory visibility, enhancing transaction reliability, supporting delivery coordination, and strengthening customer communication. These improvements contribute to better operational efficiency and provide a strong foundation for future system enhancements.

5. References

[1] “Documentation overview,” Spring.io. [Online]. Available: https://docs.spring.io/spring-boot/documentation.html. [Accessed: 21-Apr-2026].

[2] “Dependency injection,” Spring.io. [Online]. Available: https://docs.spring.io/spring-framework/reference/core/beans/dependencies/factory-collaborators.html. [Accessed: 21-Apr-2026].

[3] “Spring Boot Reference Guide,” Spring.io. [Online]. Available: https://docs.spring.io/spring-boot/docs/2.1.0.RELEASE/reference/html/index.html. [Accessed: 21-Apr-2026].

[4] “Java documentation,” Oracle Help Center, 30-July-2018. [Online]. Available: https://docs.oracle.com/en/java/index.html. [Accessed: 21-Apr-2026].

[5] “UML Class Diagram tutorial,” Visual-paradigm.com. [Online]. Available: https://www.visual-paradigm.com/guide/uml-unified-modeling-language/uml-class-diagram-tutorial/. [Accessed: 21-Apr-2026].

[6] “UML class diagram tutorial,” Lucidchart. [Online]. Available: https://www.lucidchart.com/pages/uml-class-diagram. [Accessed: 21-Apr-2026].

[7] “Enhanced ER model,” GeeksforGeeks, 24-Oct-2017. [Online]. Available: https://www.geeksforgeeks.org/dbms/enhanced-er-model/. [Accessed: 21-Apr-2026].

[8] “Manifesto for Agile Software Development,” Agilemanifesto.org. [Online]. Available: https://agilemanifesto.org/. [Accessed: 21-Apr-2026].

[9] “Introduction to software testing,” GeeksforGeeks, 08-Aug-2017. [Online]. Available: https://www.geeksforgeeks.org/software-testing/software-testing-basics/. [Accessed: 21-Apr-2026].

Appendix A: Design Diagrams

Process Models – BPMN Diagrams

- User registration

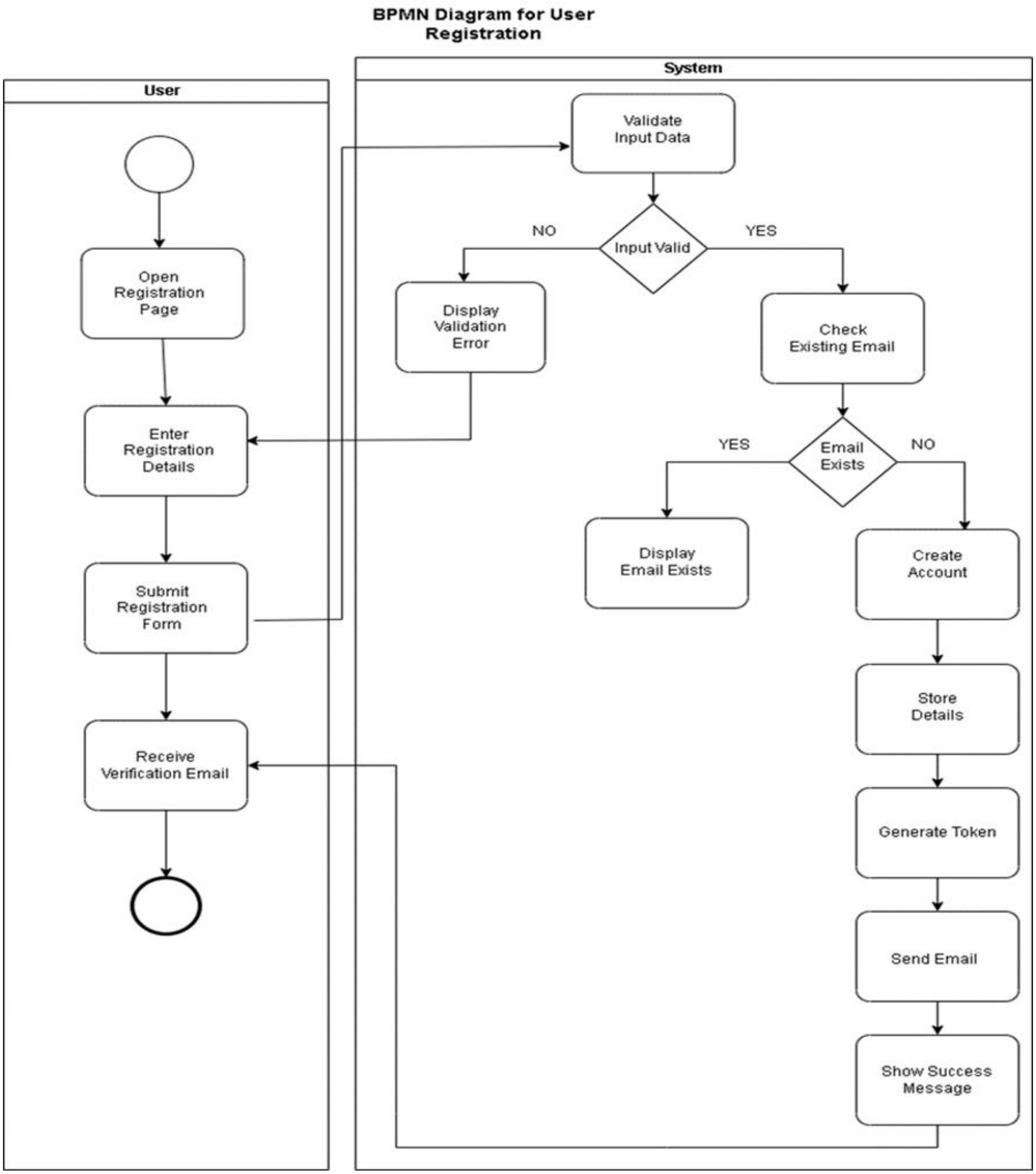

Figure 5.1 : User registration BPMN diagram

* Payment management

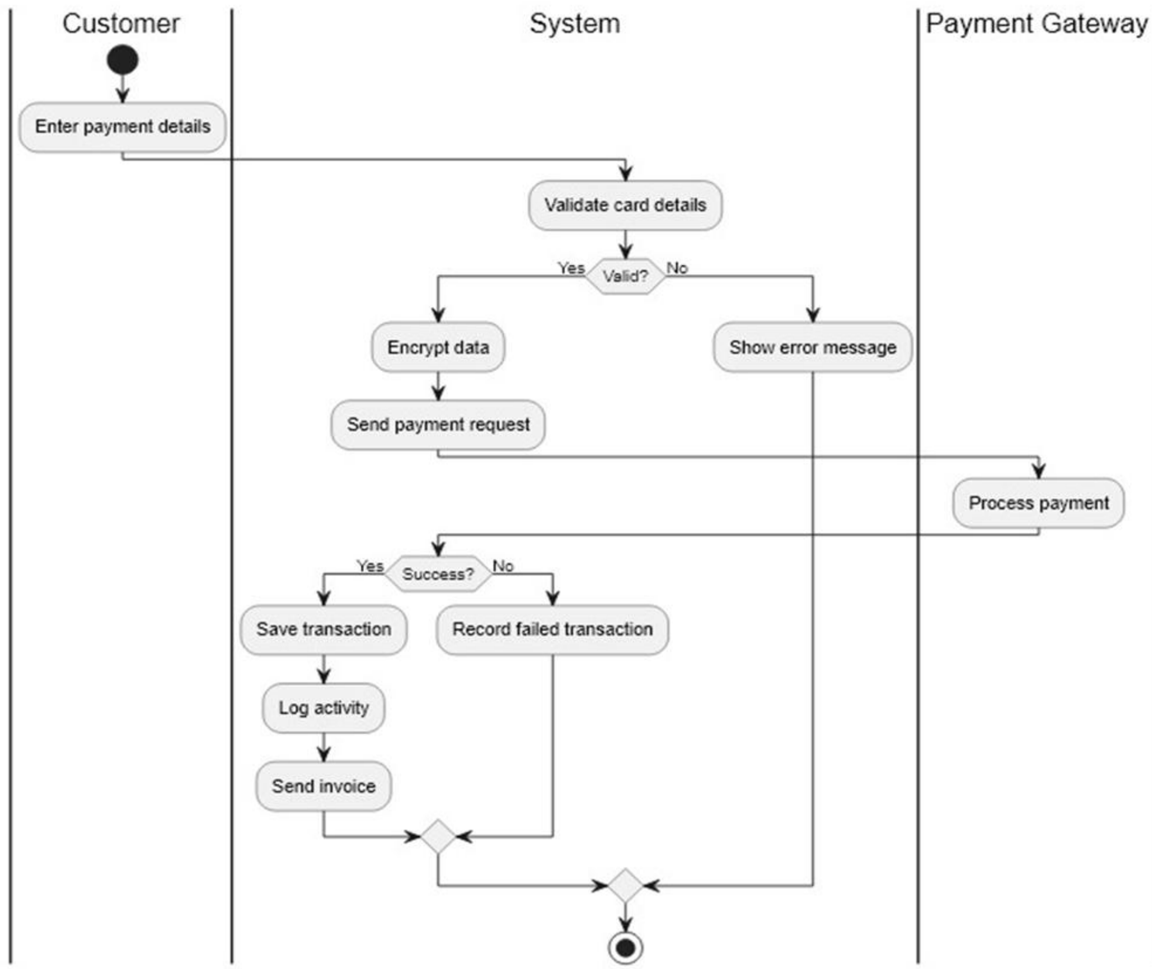

Figure 5.2 : Payment Management BPMN diagram

- Order management

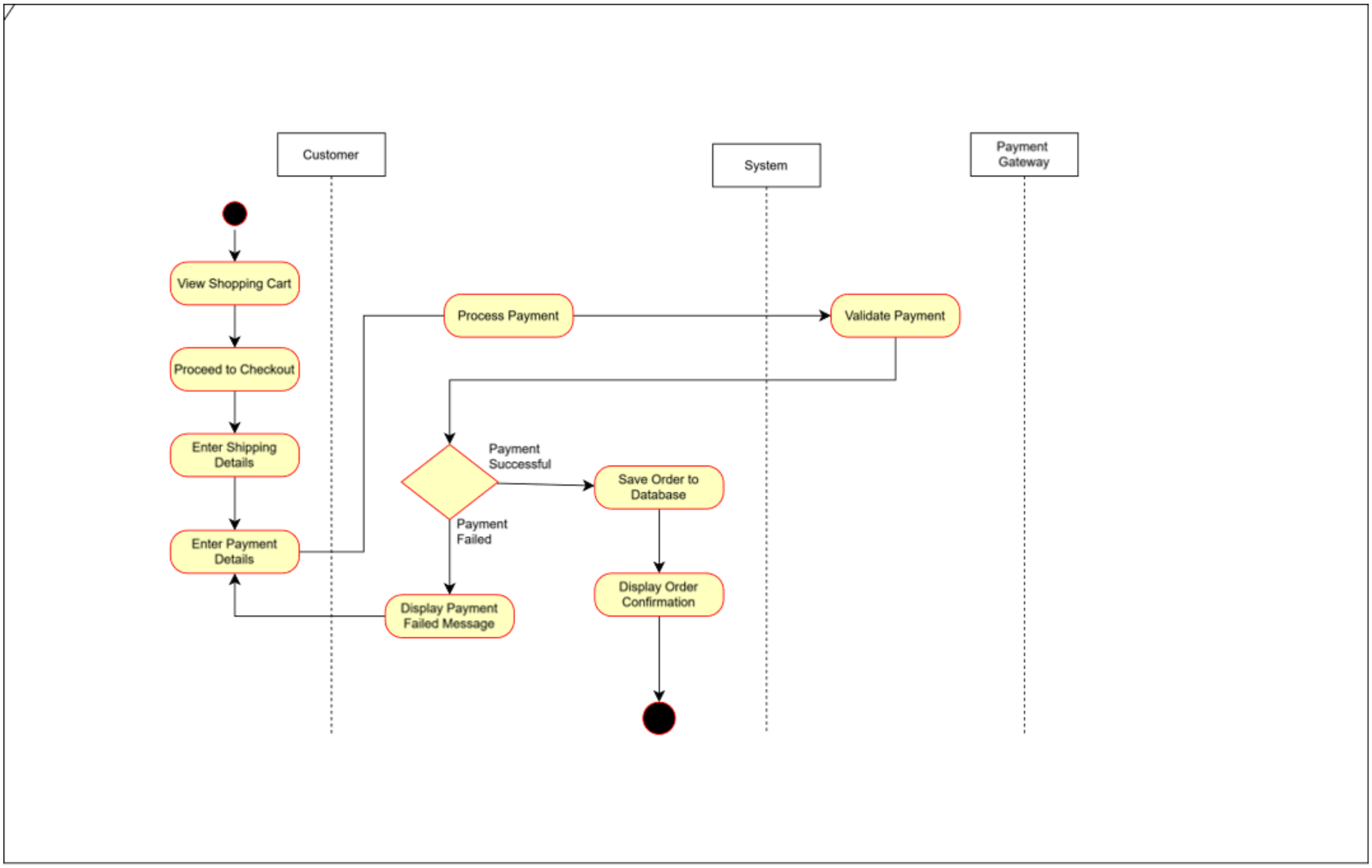

Figure 5.3 : Order Management BPMN diagram

- Delivery Management

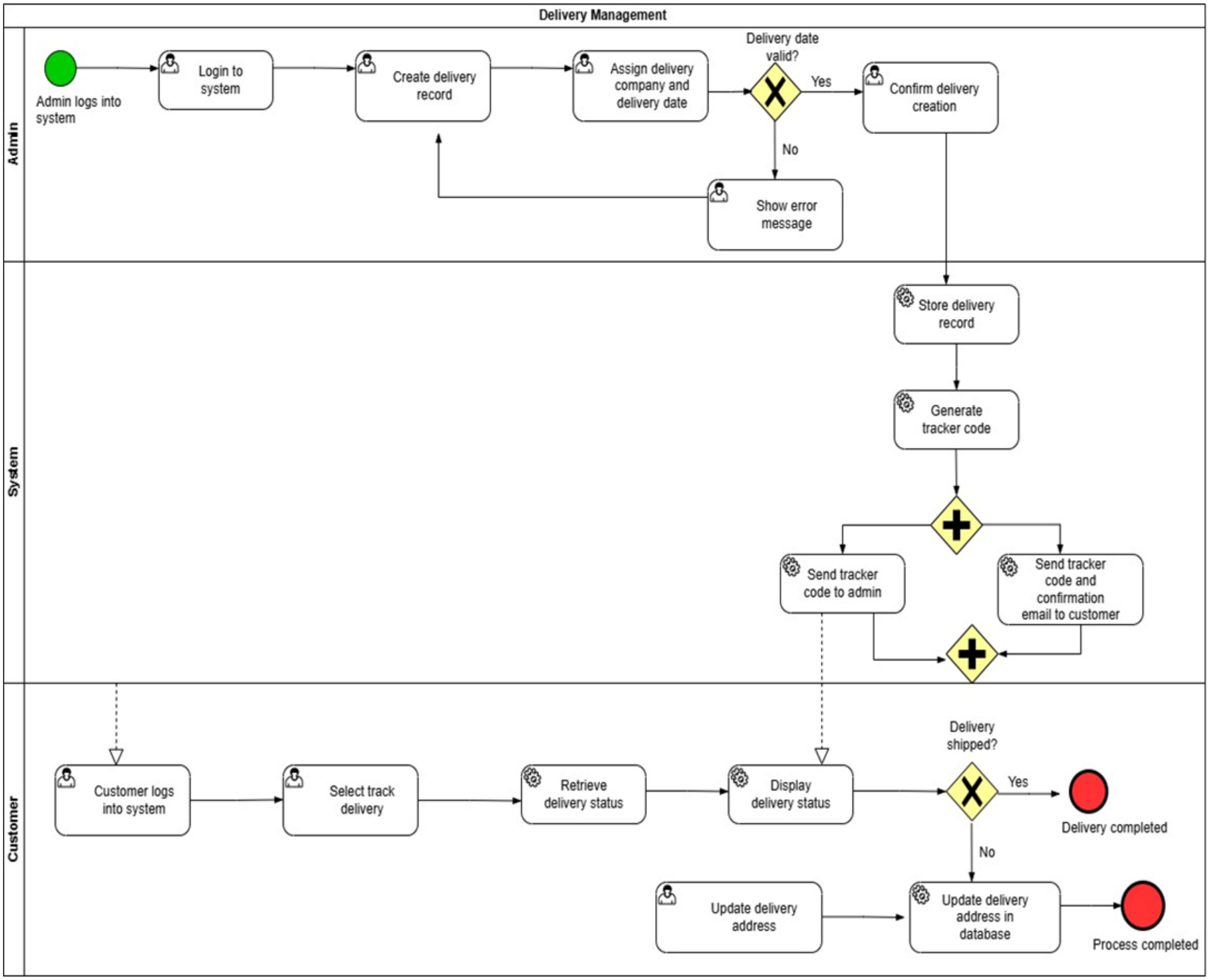

Figure 5.4 : Delivery Management BPMN diagram

Appendix B: Test Results

This appendix presents supporting evidence for the testing activities conducted for the Integrated Furniture Shop Management System. Additional verification artifacts are provided here to complement the functional testing results discussed in Section 2.4. These include representative detailed test execution records, validation of error-handling behavior, module-level coverage summaries.

Sample Detailed Functional Test Case

Test Case 1 — User Management Module

Table 5.1: Test Case 1 — User Management Module

Test case ID	TC_UM_001
Test title:	User Registration Functionality
Test priority	High
Module name:	User Management Module
Description:	Verify that a new user can successfully register using valid details.
Preconditions:	User has access to registration page.
Dependencies:	Database connection available
Test steps:	1) Navigate to the registration page 2) Enter valid name, email, and password 3) Click **Register** button 4) Verify confirmation message displayed 5) Verify verification email is sent
Expected Result:	User account created successfully and verification email sent

Test Case 2 — Product Management Module

Table 5.2: Test Case 2 — Product Management Module

Test case ID	TC_PM_001
Test title:	Add Product Functionality
Test priority	High
Module name:	Product Management Module
Description:	Verify that administrator can add a new product successfully.
Preconditions:	Administrator logged into system
Dependencies:	Database connection available
Test steps:	1) Navigate to Product Management 2) Click **Add Product** 3) Enter product name, description, category, and price 4) Upload product image 5) Click **Save Product**
Expected Result:	New product stored successfully and displayed in product list.

Test Case 3 — Shopping Cart & Order Management Module

Table 5.3: Test Case 3 — Shopping Cart & Order Management Module

Test case ID	TC_OM_001
Test title:	Test Order Placement Functionality
Test priority	High
Module name:	Shopping Cart & Order Management Module
Description:	Verify that a customer can place an order successfully.
Preconditions:	Customer logged into system Product available in stock
Dependencies:	Database connection available

Test steps:	1) Select product from product list 2) Click Add to Cart 3) Navigate to shopping cart 4) Click Checkout 5) Confirm order details
Expected Result:	Order created successfully and confirmation message displayed.

Test Case 4 — Payment Management Module

Table 5.4: Test Case 4 — Payment Management Module

Test case ID	TC_PB_001
Test title:	Test Payment Processing Functionality
Test priority	High
Module name:	Payment Management Module
Description:	Verify that customer can complete payment using valid card detail
Preconditions:	Customer has placed order
Dependencies:	Payment gateway available
Test steps:	1) Navigate to Payment page 2) Enter valid card details 3) Click Confirm Payment 4) Wait for transaction processing
Expected Result:	Payment completed successfully and invoice generated

Test Case 5 — Delivery Management Module

Table 5.5: Test Case 5 — Delivery Management Module

Test case ID	TC_DM_001
Test title:	Test Delivery Status Update Functionality
Test priority	High
Module name:	Delivery Management Module
Description:	Verify that delivery manager can update delivery status successfully.
Preconditions:	Delivery record exists

	delivery manager logged into system
Dependencies:	Database connection available
Test steps:	1) Navigate to Delivery Management section 2) Select assigned delivery record 3) Update delivery status to **In Transit** 4) Save changes
Expected Result:	Delivery status updated successfully and visible to customer.

Test Case 6 — Feedback & Support Management Module

Table 5.6: Test Case 6 — Feedback & Support Management Module

Test case ID	TC_FS_001
Test title:	Test Support Ticket Creation Functionality
Test priority	High
Module name:	Feedback & Support Management Module
Description:	Verify that customer can create support ticket successfully.
Preconditions:	Customer logged into system
Dependencies:	Database connection available
Test steps:	1) Navigate to Support section 2) Click Create Support Ticket 3) Enter issue description 4) Submit request
Expected Result:	Support ticket created successfully and ticket ID generated.

Appendix C: Selected Code Listings

Monthly Sales Report Generation Algorithm

The following code segment demonstrates the implementation of the monthly sales report generation logic used within the administrative reporting module. The algorithm filters cancelled and refunded transactions before calculating total revenue and completed order statistics.

```
for (Order order : allOrders) {

    Payment payment =
```

```
        paymentRepo.findFirstByOrder_OrderIdOrderByTransactionDateDesc(
            order.getOrderId()
        ).orElse(null);

    boolean isRefunded =
        payment != null &&
        payment.getStatus() == PaymentStatus.REFUNDED;

    if (order.getStatus() != OrderStatus.CANCELLED && !isRefunded) {

        totalSales =
            totalSales.add(order.getTotalAmount());

        totalOrders++;

        if (order.getStatus() == OrderStatus.DELIVERED ||
            order.getStatus() == OrderStatus.SHIPPED) {

            successfulOrders++;
        }
    }
}
```

This reporting algorithm improves financial accuracy by excluding cancelled and refunded orders while generating summarized administrative sales statistics used for business performance monitoring.

Invoice Email Generation with PDF Attachment Algorithm

The following code segment demonstrates the implementation of automated invoice email generation within the EmailService class. This method creates an electronic invoice in PDF format using the iText library and attaches it to a confirmation email sent to the customer after successful payment processing.

```
public void sendInvoiceEmail(User user, Order order) {
    try {
        MimeMessage message = mailSender.createMimeMessage();
        MimeMessageHelper helper = new MimeMessageHelper(message, true, "UTF-8");


        helper.setFrom(fromEmail);
        helper.setTo(user.getEmail());
        helper.setSubject("Invoice for Order #" + order.getOrderId());


        String htmlBody = """
```

```
            <h2>Thank you for your purchase!</h2>
            <p>Dear %s,</p>
            <p>Please find attached the official invoice for your recent order.</p>
            <p><strong>Order ID:</strong> %d<br>
            <strong>Total Amount:</strong> Rs. %s</p>
            """.formatted(
                user.getFirstName(),
                order.getOrderId(),
                order.getTotalAmount()
        );
        helper.setText(htmlBody, true);
        byte[] pdfBytes = generateSimpleInvoicePdf(order, user);

        helper.addAttachment(
                "Invoice_Order_" + order.getOrderId() + ".pdf",
                new ByteArrayResource(pdfBytes)
        );
        mailSender.send(message);
    } catch (Exception e) {
        System.err.println("Failed to send invoice email");
    }
}
```

Customer Feedback

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Business Reg No: W/N.8246 CRTD Reg No : TD/2708/AA

23 May 2026 Serial No:3389

Dear Development Team,

Appreciation for Anura Furniture Web System Development

We, at Anura Furniture, would like to express our sincere appreciation to the group of students from Sri Lanka Institute of Information Technology (SLIIT) for successfully contributing to the development, testing, and deployment of a web-based management system for our company as part of their academic project.

This project was carried out by a dedicated team of six students from the Information Systems Engineering (ISE) program. Throughout the project, the team demonstrated excellent teamwork, technical knowledge, commitment, and professionalism.

The developed system was designed to simplify and improve our business operations by digitalizing and managing several important activities within the company. The system includes the following major components:

- Customer Management
- Product Management
- Shopping Cart and Order Management
- Payment Management
- Delivery Management
- Feedback and Support Tickets Management

In addition to system development, the students also contributed significantly to system testing, bug fixing, deployment support, and overall system improvement to ensure the application functions smoothly and efficiently for our business environment. Their contribution helped reduce manual work, improve accuracy, and enhance the efficiency of our daily operations.

We highly appreciate the effort, creativity, and dedication shown by all six team members during the entire project lifecycle. Their ability to understand business requirements and provide practical technical solutions was truly commendable.

We wish them continued success in their academic and professional careers and thank them for their valuable contribution to Anura Furniture.

Yours faithfully,

....

Managing Director,
Anura Furniture,
Dekatana

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